# Mathematics - Class 4

Solved Exercises - Complete Book (Punjab Curriculum and Textbook Board, Lahore)

## Unit 1 - Whole Numbers (Section 1)

Complete-book coverage: Unit 1 Sections 1-3 (Whole Numbers), Unit 2 Factors and Multiples, Unit 3 Fractions, Unit 4 Decimals, Unit 5 Measurement and Time, Unit 6 Geometry, Unit 7 Data Handling. Every answer is solved in explicit steps and cross-checked against the book's own Answers section (book pages 175-181).

Method used throughout: compare and expand numbers by place value, from the left (ten-thousands, thousands, hundreds, tens, ones).

Flags carried (see manifest for full reasoning): Review Exercise Q1(d) - one printed option mis-OCR'd as '666 66', read as 99,999, the greatest 5-digit number (book p 7). Review Exercise Q4(c) - the book prints 'sixty hundred', a misprint; answered 15,660 per the book's own answer list. Review Exercise Q4(a) - answer-list OCR '51568' vs the question's 'eighty-six', answered 51,586. Review Q5 coloured digits confirmed at 300 DPI and against the answer list.

OCR-ambiguity notes (answers unchanged, arithmetic governs): Section 2 Review Q1 MCQ key letter-fragments are column-scrambled in the Answers OCR. Unit 2 Review Q1(e) key fragment reads '(e)ii' where LCM(5,10) = 10 = option (iii) - a dropped-'i' artefact.

## Unit 1 - Whole Numbers (Section 1)

Method used throughout: compare and expand numbers by place value, from the left (ten-thousands, thousands, hundreds, tens, ones).

### Exercise 1 (book page 5)

1. Write the following numbers in the expanded form: (a) 75,432

75,432 = 7 ten-thousands + 5 thousands + 4 hundreds + 3 tens + 2 ones = 70,000 + 5,000 + 400 + 30 + 2.

(b) 37,911
37,911 = 3 ten-thousands + 7 thousands + 9 hundreds + 1 ten + 1 one = 30,000 + 7,000 + 900 + 10 + 1.

(c) 10,956
10,956 = 1 ten-thousand + 0 thousands + 9 hundreds + 5 tens + 6 ones = 10,000 + 0 + 900 + 50 + 6.

(d) 46,743
46,743 = 4 ten-thousands + 6 thousands + 7 hundreds + 4 tens + 3 ones = 40,000 + 6,000 + 700 + 40 + 3.

(e) 86,594
86,594 = 8 ten-thousands + 6 thousands + 5 hundreds + 9 tens + 4 ones = 80,000 + 6,000 + 500 + 90 + 4.

(f) 19,223
19,223 = 1 ten-thousand + 9 thousands + 2 hundreds + 2 tens + 3 ones = 10,000 + 9,000 + 200 + 20 + 3.

(g) 22,167
22,167 = 2 ten-thousands + 2 thousands + 1 hundred + 6 tens + 7 ones = 20,000 + 2,000 + 100 + 60 + 7.

(h) 57,890
57,890 = 5 ten-thousands + 7 thousands + 8 hundreds + 9 tens + 0 ones = 50,000 + 7,000 + 800 + 90 + 0.

(i) 36,789
36,789 = 3 ten-thousands + 6 thousands + 7 hundreds + 8 tens + 9 ones = 30,000 + 6,000 + 700 + 80 + 9.

(j) 78,324
78,324 = 7 ten-thousands + 8 thousands + 3 hundreds + 2 tens + 4 ones = 70,000 + 8,000 + 300 + 20 + 4.

(k) 41,452
41,452 = 4 ten-thousands + 1 thousand + 4 hundreds + 5 tens + 2 ones = 40,000 + 1,000 + 400 + 50 + 2.

(l) 56,432
56,432 = 5 ten-thousands + 6 thousands + 4 hundreds + 3 tens + 2 ones = 50,000 + 6,000 + 400 + 30 + 2.

2. Write the following numbers given in their expanded form into standard form: (a) 20,000 + 1,000 + 100 + 70 + 1

20,000 + 1,000 + 100 + 70 + 1 = 21,171.

(b) 30,000 + 9,000 + 200 + 30 + 5

30,000 + 9,000 + 200 + 30 + 5 = 39,235.

(c) 60,000 + 5,000 + 300 + 40 + 3

60,000 + 5,000 + 300 + 40 + 3 = 65,343.

(d) 50,000 + 6,000 + 700 + 90 + 0

50,000 + 6,000 + 700 + 90 + 0 = 56,790.

3. Write the place and place value of the coloured digits: (a) 76,102 (coloured digit 0)

The coloured digit is 0. It is at the tens place. Its place value = 0 x 10 = 0.

(b) 24,360 (coloured digit 2)

The coloured digit is 2. It is at the ten-thousands place. Its place value = 2 x 10,000 = 20,000.

(c) 94,615 (coloured digit 4)

The coloured digit is 4. It is at the thousands place. Its place value = 4 x 1,000 = 4,000.

(d) 65,496 (coloured digit 6)

The coloured digit is 6. It is at the ones place. Its place value = 6 x 1 = 6.

(e) 73,456 (coloured digit 7)

The coloured digit is 7. It is at the ten-thousands place. Its place value = 7 x 10,000 = 70,000.

(f) 18,654 (coloured digit 8)

The coloured digit is 8. It is at the thousands place. Its place value = 8 x 1,000 = 8,000.

(g) 34,566 (coloured digit 5)

The coloured digit is 5. It is at the hundreds place. Its place value = 5 x 100 = 500.

(h) 86,042 (coloured digit 6)

The coloured digit is 6. It is at the thousands place. Its place value = 6 x 1,000 = 6,000.

(i) 56,324 (coloured digit 4)

The coloured digit is 4. It is at the ones place. Its place value = 4 x 1 = 4.

4. Write the following numbers in words: (a) 74,325

74,325 = 74 thousand + 325 = seventy-four thousand three hundred twenty-five.

(b) 43,711

43,711 = 43 thousand + 711 = forty-three thousand seven hundred eleven.

(c) 19,560

19,560 = 19 thousand + 560 = nineteen thousand five hundred sixty.

(d) 75,434

75,434 = 75 thousand + 434 = seventy-five thousand four hundred thirty-four.

(e) 67,459

67,459 = 67 thousand + 459 = sixty-seven thousand four hundred fifty-nine.

(f) 25,302

25,302 = 25 thousand + 302 = twenty-five thousand three hundred two.

(g) 36,721

36,721 = 36 thousand + 721 = thirty-six thousand seven hundred twenty-one.

(h) 78,065

78,065 = 78 thousand + 65 = seventy-eight thousand sixty-five.

(i) 62,897

62,897 = 62 thousand + 897 = sixty-two thousand eight hundred ninety-seven.

(j) 37,264

37,264 = 37 thousand + 264 = thirty-seven thousand two hundred sixty-four.

(k) 45,129

45,129 = 45 thousand + 129 = forty-five thousand one hundred twenty-nine.

(l) 43,275

43,275 = 43 thousand + 275 = forty-three thousand two hundred seventy-five.

5. Write the following in numerals: (a) Twenty-five thousand six hundred

Twenty-five thousand six hundred = 25 thousand + 600 = 25,600.

(b) Seventy-eight thousand four hundred two

Seventy-eight thousand four hundred two = 78 thousand + 402 = 78,402.

(c) Forty-one thousand sixty-one

Forty-one thousand sixty-one = 41 thousand + 61 = 41,061.

(d) Ninety-eight thousand three hundred one

Ninety-eight thousand three hundred one = 98 thousand + 301 = 98,301.

(e) Seventy-two thousand five hundred forty-six

Seventy-two thousand five hundred forty-six = 72 thousand + 546 = 72,546.

(f) Twelve thousand five hundred fifty-five

Twelve thousand five hundred fifty-five = 12 thousand + 555 = 12,555.

(g) Ninety-eight thousand five

Ninety-eight thousand five = 98 thousand + 5 = 98,005.

(h) Forty-eight thousand four hundred forty-four

Forty-eight thousand four hundred forty-four = 48 thousand + 444 = 48,444.

(i) Eighty-eight thousand three hundred twenty

Eighty-eight thousand three hundred twenty = 88 thousand + 320 = 88,320.

## Exercise 2 (book pages 8-9)

1. Compare the following numbers by using symbols (<, >, =): (a) 84,325 ___ 93,417

84,325 < 93,417. Compare from the left: at the ten-thousands place 8 < 9, so 84,325 is smaller.

(b) 4,853 ___ 19,314

4,853 < 19,314. A 4-digit number is always smaller than a 5-digit number.

(c) 56,708 ___ 32,156

56,708 > 32,156. At the ten-thousands place 5 > 3, so 56,708 is greater.

(d) 23,612 ___ 23,612
23,612 = 23,612. Both numbers are exactly the same.

(e) 65,356 ___ 65,358
65,356 < 65,358. Ten-thousands, thousands, hundreds and tens are equal; at the ones place 6 < 8.

(f) 74,932 ___ 74,542
74,932 > 74,542. Ten-thousands and thousands are equal; at the hundreds place 9 > 5.

(g) 68,709 ___ 68,709
68,709 = 68,709. Both numbers are exactly the same.

(h) 32,567 ___ 23,578
32,567 > 23,578. At the ten-thousands place 3 > 2, so 32,567 is greater.

2. Write the following numbers in an ascending order: (a) 40,131; 40,735; 31,273
Ascending order: 31,273; 40,131; 40,735. (31,273 is smallest; then 40,131 < 40,735.)

(b) 30,817; 28,211; 43,181
Ascending order: 28,211; 30,817; 43,181.

(c) 70,442; 58,375; 84,176
Ascending order: 58,375; 70,442; 84,176.

(d) 67,319; 22,342; 97,323
Ascending order: 22,342; 67,319; 97,323.

(e) 83,624; 36,241; 63,283
Ascending order: 36,241; 63,283; 83,624.

(f) 48,326; 23,634; 43,124
Ascending order: 23,634; 43,124; 48,326.

(g) 59,312; 60,337; 24,085
Ascending order: 24,085; 59,312; 60,337.

(h) 89,675; 84,675; 89,546
Ascending order: 84,675; 89,546; 89,675. (89,546 < 89,675 because at the

tens place 4 < 7.)

3. Write the following numbers in descending order: (a) 83,401; 97,035; 12,337
Descending order: 97,035; 83,401; 12,337.

(b) 18,017; 18,221; 13,411
Descending order: 18,221; 18,017; 13,411.

(c) 42,734; 53,358; 48,176
Descending order: 53,358; 48,176; 42,734.

(d) 36,121; 34,222; 37,923
Descending order: 37,923; 36,121; 34,222.

(e) 16,483; 23,601; 36,243
Descending order: 36,243; 23,601; 16,483.

(f) 12,683; 24,313; 24,391
Descending order: 24,391; 24,313; 12,683.

(g) 32,531; 36,537; 28,540
Descending order: 36,537; 32,531; 28,540.

(h) 98,754; 78,543; 89,654
Descending order: 98,754; 89,654; 78,543.

## Review Exercise (book pages 10-12)

1. Choose the correct option: (a) The smallest 6-digit number is:
(a) 100,000 ✓
(b) 111,111
(c) 101,010
(d) 111,000

(b) Comparison of numbers always starts from the:
(a) right
(b) left ✓
(c) last
(d) above

(c) In number 38,101, the place value of digit 8 is:
(a) 8
(b) 800
(c) 8,000 ✓

(d) 80

(d) The greatest 5-digit number is:
(a) 99,999
(b) 91,100
(c) 90,000
(d) 90,101

(e) 34,011 is greater than:
(a) 34,010
(b) 34,212
(c) 34,210
(d) 34,111

(f) 31,108 is smaller than:
(a) 31,106
(b) 30,100
(c) 31,107
(d) 31,109

Working for (a): the smallest 6-digit number is 100,000 (99,999 + 1). Working for (c): in 38,101 the digit 8 is at the thousands place, so its place value is 8 x 1,000 = 8,000. Working for (d): the book itself teaches (page 7) that 99,999 is the greatest 5-digit whole number. Working for (e): only 34,010 is smaller than 34,011. Working for (f): only 31,109 is greater than 31,108.

2. Write the following numbers in words: (a) 43,567
43,567 = 43 thousand + 567 = forty-three thousand five hundred sixty-seven.

(b) 97,741
97,741 = 97 thousand + 741 = ninety-seven thousand seven hundred forty-one.

(c) 52,016
52,016 = 52 thousand + 16 = fifty-two thousand sixteen.

(d) 46,743
46,743 = 46 thousand + 743 = forty-six thousand seven hundred forty-three.

(e) 58,649
58,649 = 58 thousand + 649 = fifty-eight thousand six hundred forty-nine.

(f) 95,202
95,202 = 95 thousand + 202 = ninety-five thousand two hundred two.

(g) 10,007
10,007 = 10 thousand + 7 = ten thousand seven.

(h) 86,950
86,950 = 86 thousand + 950 = eighty-six thousand nine hundred fifty.

(i) 60,000
60,000 = 60 thousand = sixty thousand.

(j) 60,032
60,032 = 60 thousand + 32 = sixty thousand thirty-two.

(k) 52,901
52,901 = 52 thousand + 901 = fifty-two thousand nine hundred one.

(l) 36,427
36,427 = 36 thousand + 427 = thirty-six thousand four hundred twenty-seven.

3. Write the following numbers in the expanded form: (a) 53,672
53,672 = 50,000 + 3,000 + 600 + 70 + 2.

(b) 74,311
74,311 = 70,000 + 4,000 + 300 + 10 + 1.

(c) 25,609
25,609 = 20,000 + 5,000 + 600 + 0 + 9.

(d) 65,743
65,743 = 60,000 + 5,000 + 700 + 40 + 3.

(e) 78,945
78,945 = 70,000 + 8,000 + 900 + 40 + 5.

(f) 92,502
92,502 = 90,000 + 2,000 + 500 + 0 + 2.

(g) 16,272
16,272 = 10,000 + 6,000 + 200 + 70 + 2.

(h) 69,078

69,078 = 60,000 + 9,000 + 0 + 70 + 8.

(i) 27,869
27,869 = 20,000 + 7,000 + 800 + 60 + 9.

(j) 32,786
32,786 = 30,000 + 2,000 + 700 + 80 + 6.

(k) 41,902
41,902 = 40,000 + 1,000 + 900 + 0 + 2.

(l) 64,753
64,753 = 60,000 + 4,000 + 700 + 50 + 3.

4. Write the following in numerals: (a) Fifty-one thousand five hundred eighty-six
Fifty-one thousand five hundred eighty-six = 51 thousand + 586 = 51,586.

(b) Eighty-two thousand four hundred four
Eighty-two thousand four hundred four = 82 thousand + 404 = 82,404.

(c) Fifteen thousand sixty hundred sixty (as printed in the book)
The book's own answer list gives 15,660 for this item, i.e. the intended reading is fifteen thousand six hundred sixty = 15 thousand + 660 = 15,660. (The printed wording 'sixty hundred' is a misprint for 'six hundred'; the answer follows the book's answer list.)

(d) Twenty-one thousand one hundred five
Twenty-one thousand one hundred five = 21 thousand + 105 = 21,105.

(e) Twenty-three thousand five hundred six
Twenty-three thousand five hundred six = 23 thousand + 506 = 23,506.

(f) Ninety-six thousand one hundred twenty-five
Ninety-six thousand one hundred twenty-five = 96 thousand + 125 = 96,125.

(g) Sixty-seven thousand three
Sixty-seven thousand three = 67 thousand + 3 = 67,003.

5. Write the place and place value of the coloured digits: (a) 65,021 (coloured digit 5)
The coloured digit is 5. It is at the thousands place. Its place value = 5 x 1,000 = 5,000.

(b) 24,360 (coloured digit 2)
The coloured digit is 2. It is at the ten-thousands place. Its place value = 2 x 10,000 = 20,000.

(c) 46,715 (coloured digit 4)
The coloured digit is 4. It is at the ten-thousands place. Its place value = 4 x 10,000 = 40,000.

(d) 60,704 (coloured digit 0)
The coloured digit is 0. It is at the tens place. Its place value = 0 x 10 = 0.

(e) 57,564 (coloured digit 4)
The coloured digit is 4. It is at the ones place. Its place value = 4 x 1 = 4.

(f) 26,514 (coloured digit 6)
The coloured digit is 6. It is at the thousands place. Its place value = 6 x 1,000 = 6,000.

(g) 52,663 (coloured digit 5)
The coloured digit is 5. It is at the ten-thousands place. Its place value = 5 x 10,000 = 50,000.

(h) 34,540 (coloured digit 3)
The coloured digit is 3. It is at the ten-thousands place. Its place value = 3 x 10,000 = 30,000.

6. Write the following in standard form: (a) 40,000 + 4,000 + 600 + 80 + 3
40,000 + 4,000 + 600 + 80 + 3 = 44,683.

(b) 90,000 + 0,000 + 000 + 50 + 4
90,000 + 0,000 + 000 + 50 + 4 = 90,054.

(c) 20,000 + 9,000 + 100 + 00 + 4
20,000 + 9,000 + 100 + 00 + 4 = 29,104.

(d) 10,000 + 6,000 + 700 + 80 + 5
10,000 + 6,000 + 700 + 80 + 5 = 16,785.

7. Compare the following numbers by using symbols (<, >, =): (a) 5,847 ___ 31,341
5,847 < 31,341. A 4-digit number is always smaller than a 5-digit number.

(b) 34,875 ___ 98,317

34,875 < 98,317. At the ten-thousands place 3 < 9.

(c) 50,678 ___ 45,321
50,678 > 45,321. At the ten-thousands place 5 > 4.

(d) 75,326 ___ 21,635
75,326 > 21,635. At the ten-thousands place 7 > 2.

(e) 76,643 ___ 76,643
76,643 = 76,643. Both numbers are exactly the same.

(f) 37,256 ___ 54,490
37,256 < 54,490. At the ten-thousands place 3 < 5.

(g) 24,351 ___ 32,674
24,351 < 32,674. At the ten-thousands place 2 < 3.

(h) 66,809 ___ 26,228
66,809 > 26,228. At the ten-thousands place 6 > 2.

8. Write the following numbers in an ascending order: (a) 94,041; 84,405; 33,731
Ascending order: 33,731; 84,405; 94,041.

(b) 19,375; 12,921; 14,131
Ascending order: 12,921; 14,131; 19,375.

(c) 45,034; 37,358; 42,876
Ascending order: 37,358; 42,876; 45,034.

(d) 36,172; 35,242; 37,723
Ascending order: 35,242; 36,172; 37,723.

9. Write the following numbers in descending order: (a) 12,683; 14,601; 18,624
Descending order: 18,624; 14,601; 12,683.

(b) 16,283; 26,133; 14,394
Descending order: 26,133; 16,283; 14,394.

(c) 23,913; 30,536; 22,480
Descending order: 30,536; 23,913; 22,480.

(d) 54,788; 54,786; 54,790

Descending order: 54,790; 54,788; 54,786. (All three agree up to the tens place; at the ones place 0 > 8 > 6.)

## Unit 1 - Addition and Subtraction (Section 2)

### Exercise 1 (book page 16)

Method: add column by column from the ones place, carrying to the next place whenever a column total is 10 or more.

1. (a) 13,927 + 2,415

Ones place: 7 + 5 = 12. Write 2 and carry 1.

Tens place: 2 + 1 + 1 (carry) = 4. Write 4.

Hundreds place: 9 + 4 = 13. Write 3 and carry 1.

Thousands place: 3 + 2 + 1 (carry) = 6. Write 6.

Ten-thousands place: 1 + 0 = 1. Write 1.

**Answer: 16,342.**

1. (b) 35,321 + 4,649

Ones place: 1 + 9 = 10. Write 0 and carry 1.

Tens place: 2 + 4 + 1 (carry) = 7. Write 7.

Hundreds place: 3 + 6 = 9. Write 9.

Thousands place: 5 + 4 = 9. Write 9.

Ten-thousands place: 3 + 0 = 3. Write 3.

**Answer: 39,970.**

1. (c) 75,679 + 16,828

Ones place: 9 + 8 = 17. Write 7 and carry 1.

Tens place: 7 + 2 + 1 (carry) = 10. Write 0 and carry 1.

Hundreds place: 6 + 8 + 1 (carry) = 15. Write 5 and carry 1.

Thousands place: 5 + 6 + 1 (carry) = 12. Write 2 and carry 1.

Ten-thousands place: 7 + 1 + 1 (carry) = 9. Write 9.

**Answer: 92,507.**

1. (d) 35,312 + 24,568
Ones place: 2 + 8 = 10. Write 0 and carry 1.

Tens place: 1 + 6 + 1 (carry) = 8. Write 8.

Hundreds place: 3 + 5 = 8. Write 8.

Thousands place: 5 + 4 = 9. Write 9.

Ten-thousands place: 3 + 2 = 5. Write 5.

**Answer: 59,880.**

2. (a) 58,134 + 45,367
Ones place: 4 + 7 = 11. Write 1 and carry 1.

Tens place: 3 + 6 + 1 (carry) = 10. Write 0 and carry 1.

Hundreds place: 1 + 3 + 1 (carry) = 5. Write 5.

Thousands place: 8 + 5 = 13. Write 3 and carry 1.

Ten-thousands place: 5 + 4 + 1 (carry) = 10. Write 0 and carry 1.

Ten-thousands carry: write 1.

**Answer: 103,501.**

2. (b) 78,954 + 12,236
Ones place: 4 + 6 = 10. Write 0 and carry 1.

Tens place: 5 + 3 + 1 (carry) = 9. Write 9.

Hundreds place: 9 + 2 = 11. Write 1 and carry 1.

Thousands place: 8 + 2 + 1 (carry) = 11. Write 1 and carry 1.

Ten-thousands place: 7 + 1 + 1 (carry) = 9. Write 9.

**Answer: 91,190.**

2. (c) 89,764 + 97,856
Ones place: 4 + 6 = 10. Write 0 and carry 1.

Tens place: 6 + 5 + 1 (carry) = 12. Write 2 and carry 1.

Hundreds place: 7 + 8 + 1 (carry) = 16. Write 6 and carry 1.

Thousands place: 9 + 7 + 1 (carry) = 17. Write 7 and carry 1.

Ten-thousands place: 8 + 9 + 1 (carry) = 18. Write 8 and carry 1.

Ten-thousands carry: write 1.

**Answer: 187,620.**

2. (d) 53,241 + 67,543
Ones place: 1 + 3 = 4. Write 4.

Tens place: 4 + 4 = 8. Write 8.

Hundreds place: 2 + 5 = 7. Write 7.

Thousands place: 3 + 7 = 10. Write 0 and carry 1.

Ten-thousands place: 5 + 6 + 1 (carry) = 12. Write 2 and carry 1.

Ten-thousands carry: write 1.

**Answer: 120,784.**

2. (e) 98,756 + 56,744
Ones place: 6 + 4 = 10. Write 0 and carry 1.

Tens place: 5 + 4 + 1 (carry) = 10. Write 0 and carry 1.

Hundreds place: 7 + 7 + 1 (carry) = 15. Write 5 and carry 1.

Thousands place: 8 + 6 + 1 (carry) = 15. Write 5 and carry 1.

Ten-thousands place: 9 + 5 + 1 (carry) = 15. Write 5 and carry 1.

Ten-thousands carry: write 1.

**Answer: 155,500.**

2. (f) 65,432 + 87,643

Ones place: 2 + 3 = 5. Write 5.

Tens place: 3 + 4 = 7. Write 7.

Hundreds place: 4 + 6 = 10. Write 0 and carry 1.

Thousands place: 5 + 7 + 1 (carry) = 13. Write 3 and carry 1.

Ten-thousands place: 6 + 8 + 1 (carry) = 15. Write 5 and carry 1.

Ten-thousands carry: write 1.

**Answer: 153,075.**

2. (g) 42,115 + 61,537

Ones place: 5 + 7 = 12. Write 2 and carry 1.

Tens place: 1 + 3 + 1 (carry) = 5. Write 5.

Hundreds place: 1 + 5 = 6. Write 6.

Thousands place: 2 + 1 = 3. Write 3.

Ten-thousands place: 4 + 6 = 10. Write 0 and carry 1.

Ten-thousands carry: write 1.

**Answer: 103,652.**

2. (h) 58,764 + 65,744

Ones place: 4 + 4 = 8. Write 8.

Tens place: 6 + 4 = 10. Write 0 and carry 1.

Hundreds place: 7 + 7 + 1 (carry) = 15. Write 5 and carry 1.

Thousands place: 8 + 5 + 1 (carry) = 14. Write 4 and carry 1.

Ten-thousands place: 5 + 6 + 1 (carry) = 12. Write 2 and carry 1.

Ten-thousands carry: write 1.

**Answer: 124,508.**

2. (i) 54,312 + 68,534
Ones place: 2 + 4 = 6. Write 6.

Tens place: 1 + 3 = 4. Write 4.

Hundreds place: 3 + 5 = 8. Write 8.

Thousands place: 4 + 8 = 12. Write 2 and carry 1.

Ten-thousands place: 5 + 6 + 1 (carry) = 12. Write 2 and carry 1.

Ten-thousands carry: write 1.

**Answer: 122,846.**

3. Nida bought a laptop for Rs 59,453 and spent Rs 12,652 on repairing. How much total amount did she spend?
Add the two amounts: 59,453 + 12,652. Ones: 3 + 2 = 5. Tens: 5 + 5 = 10, write 0, carry 1. Hundreds: 4 + 6 + 1 = 11, write 1, carry 1. Thousands: 9 + 2 + 1 = 12, write 2, carry 1. Ten-thousands: 5 + 1 + 1 = 7. Total = Rs 72,105.

4. In January, 83,215 people travelled and in February, 21,084 people travelled from an airport. How many passengers travelled in two months?
Add: 83,215 + 21,084 = 104,299. Ones: 5 + 4 = 9. Tens: 1 + 8 = 9. Hundreds: 2 + 0 = 2. Thousands: 3 + 1 = 4. Ten-thousands: 8 + 2 = 10, write 0, carry 1 to make 104,299. So 104,299 passengers travelled in two months.

5. (a) In a library, there are 42,725 books. Administration decided to add 22,500 new books. Find the total number of books in the library.
42,725 + 22,500 = 65,225 books.

5. (b) If 23,890 more books are added then find the total number of books.
65,225 + 23,890 = 89,115 books.

6. (a) A bus covered 23,672 km distance in the first month, in the second month the same bus covered a distance of 31,716 km. Find the total distance covered in two months.
23,672 + 31,716 = 55,388 km.

6. (b) In which month did it cover more distance?

Second month: 31,716 km is more than 23,672 km, so the bus covered more distance in the second month.

## Exercise 2 (book page 19)

Method: subtract column by column from the ones place, borrowing from the next place whenever the upper digit is smaller.

1. (a) 43,562 - 7,366

Ones place: 2 is less than 6, so borrow 1 from the next place: 12 - 6 = 6. Write 6.

Tens place: 6 - 1 (borrow) = 5, which is less than 6, so borrow 1 from the next place: 15 - 6 = 9. Write 9.

Hundreds place: 5 - 1 (borrow) = 4, then 4 - 3 = 1. Write 1.

Thousands place: 3 is less than 7, so borrow 1 from the next place: 13 - 7 = 6. Write 6.

Ten-thousands place: 4 - 1 (borrow) = 3, then 3 - 0 = 3. Write 3.

**Answer: 36,196.**

1. (b) 56,848 - 34,389

Ones place: 8 is less than 9, so borrow 1 from the next place: 18 - 9 = 9. Write 9.

Tens place: 4 - 1 (borrow) = 3, which is less than 8, so borrow 1 from the next place: 13 - 8 = 5. Write 5.

Hundreds place: 8 - 1 (borrow) = 7, then 7 - 3 = 4. Write 4.

Thousands place: 6 - 4 = 2. Write 2.

Ten-thousands place: 5 - 3 = 2. Write 2.

**Answer: 22,459.**

1. (c) 93,921 - 24,618

Ones place: 1 is less than 8, so borrow 1 from the next place: 11 - 8 = 3. Write 3.

Tens place: 2 - 1 (borrow) = 1, then 1 - 1 = 0. Write 0.

Hundreds place: 9 - 6 = 3. Write 3.

Thousands place: 3 is less than 4, so borrow 1 from the next place: 13 - 4 = 9. Write 9.

Ten-thousands place: 9 - 1 (borrow) = 8, then 8 - 2 = 6. Write 6.

**Answer: 69,303.**

1. (d) 65,675 - 16,447
Ones place: 5 is less than 7, so borrow 1 from the next place: 15 - 7 = 8. Write 8.

Tens place: 7 - 1 (borrow) = 6, then 6 - 4 = 2. Write 2.

Hundreds place: 6 - 4 = 2. Write 2.

Thousands place: 5 is less than 6, so borrow 1 from the next place: 15 - 6 = 9. Write 9.

Ten-thousands place: 6 - 1 (borrow) = 5, then 5 - 1 = 4. Write 4.

**Answer: 49,228.**

1. (e) 53,521 - 34,649
Ones place: 1 is less than 9, so borrow 1 from the next place: 11 - 9 = 2. Write 2.

Tens place: 2 - 1 (borrow) = 1, which is less than 4, so borrow 1 from the next place: 11 - 4 = 7. Write 7.

Hundreds place: 5 - 1 (borrow) = 4, which is less than 6, so borrow 1 from the next place: 14 - 6 = 8. Write 8.

Thousands place: 3 - 1 (borrow) = 2, which is less than 4, so borrow 1 from the next place: 12 - 4 = 8. Write 8.

Ten-thousands place: 5 - 1 (borrow) = 4, then 4 - 3 = 1. Write 1.

**Answer: 18,872.**

1. (f) 33,275 - 26,238
Ones place: 5 is less than 8, so borrow 1 from the next place: 15 - 8 = 7.

Write 7.

Tens place: 7 - 1 (borrow) = 6, then 6 - 3 = 3. Write 3.

Hundreds place: 2 - 2 = 0. Write 0.

Thousands place: 3 is less than 6, so borrow 1 from the next place: 13 - 6 = 7. Write 7.

Ten-thousands place: 3 - 1 (borrow) = 2, then 2 - 2 = 0. Write 0.

**Answer: 7,037.**

1. (g) 96,237 - 73,455
Ones place: 7 - 5 = 2. Write 2.

Tens place: 3 is less than 5, so borrow 1 from the next place: 13 - 5 = 8. Write 8.

Hundreds place: 2 - 1 (borrow) = 1, which is less than 4, so borrow 1 from the next place: 11 - 4 = 7. Write 7.

Thousands place: 6 - 1 (borrow) = 5, then 5 - 3 = 2. Write 2.

Ten-thousands place: 9 - 7 = 2. Write 2.

**Answer: 22,782.**

1. (h) 67,453 - 32,514
Ones place: 3 is less than 4, so borrow 1 from the next place: 13 - 4 = 9. Write 9.

Tens place: 5 - 1 (borrow) = 4, then 4 - 1 = 3. Write 3.

Hundreds place: 4 is less than 5, so borrow 1 from the next place: 14 - 5 = 9. Write 9.

Thousands place: 7 - 1 (borrow) = 6, then 6 - 2 = 4. Write 4.

Ten-thousands place: 6 - 3 = 3. Write 3.

**Answer: 34,939.**

2. (a) 45,158 - 34,756

Ones place: 8 - 6 = 2. Write 2.

Tens place: 5 - 5 = 0. Write 0.

Hundreds place: 1 is less than 7, so borrow 1 from the next place: 11 - 7 = 4. Write 4.

Thousands place: 5 - 1 (borrow) = 4, then 4 - 4 = 0. Write 0.

Ten-thousands place: 4 - 3 = 1. Write 1.

**Answer: 10,402.**

2. (b) 97,843 - 61,732
Ones place: 3 - 2 = 1. Write 1.

Tens place: 4 - 3 = 1. Write 1.

Hundreds place: 8 - 7 = 1. Write 1.

Thousands place: 7 - 1 = 6. Write 6.

Ten-thousands place: 9 - 6 = 3. Write 3.

**Answer: 36,111.**

2. (c) 99,754 - 67,584
Ones place: 4 - 4 = 0. Write 0.

Tens place: 5 is less than 8, so borrow 1 from the next place: 15 - 8 = 7. Write 7.

Hundreds place: 7 - 1 (borrow) = 6, then 6 - 5 = 1. Write 1.

Thousands place: 9 - 7 = 2. Write 2.

Ten-thousands place: 9 - 6 = 3. Write 3.

**Answer: 32,170.**

2. (d) 25,341 - 16,753
Ones place: 1 is less than 3, so borrow 1 from the next place: 11 - 3 = 8. Write 8.

Tens place: 4 - 1 (borrow) = 3, which is less than 5, so borrow 1 from the next place: 13 - 5 = 8. Write 8.

Hundreds place: 3 - 1 (borrow) = 2, which is less than 7, so borrow 1 from the next place: 12 - 7 = 5. Write 5.

Thousands place: 5 - 1 (borrow) = 4, which is less than 6, so borrow 1 from the next place: 14 - 6 = 8. Write 8.

Ten-thousands place: 2 - 1 (borrow) = 1, then 1 - 1 = 0. Write 0.

**Answer: 8,588.**

2. (e) 85,964 - 74,544
Ones place: 4 - 4 = 0. Write 0.

Tens place: 6 - 4 = 2. Write 2.

Hundreds place: 9 - 5 = 4. Write 4.

Thousands place: 5 - 4 = 1. Write 1.

Ten-thousands place: 8 - 7 = 1. Write 1.

**Answer: 11,420.**

2. (f) 63,541 - 58,463
Ones place: 1 is less than 3, so borrow 1 from the next place: 11 - 3 = 8. Write 8.

Tens place: 4 - 1 (borrow) = 3, which is less than 6, so borrow 1 from the next place: 13 - 6 = 7. Write 7.

Hundreds place: 5 - 1 (borrow) = 4, then 4 - 4 = 0. Write 0.

Thousands place: 3 is less than 8, so borrow 1 from the next place: 13 - 8 = 5. Write 5.

Ten-thousands place: 6 - 1 (borrow) = 5, then 5 - 5 = 0. Write 0.

**Answer: 5,078.**

3. (a) Saad had Rs 52,490. He bought a bicycle for Rs 15,873. How much money was left with him?

52,490 - 15,873 = Rs 36,617.

3. (b) If the price of the bicycle is Rs 18,759, then how much money will be left?
52,490 - 18,759 = Rs 33,731.

4. In a godown, there are 66,375 bags of wheat and rice. If the number of wheat bags are 44,468 then find out the number of rice bags.
66,375 - 44,468 = 21,907 rice bags.

5. The students of class 3 collected Rs 35,278 for a welfare institution while the students of class 4 collected Rs 32,184. How much more amount collected by class 3 than class 4?
35,278 - 32,184 = Rs 3,094 more collected by class 3.

6. A candidate got 62,436 votes from one constituency while the other candidate got 86,733 votes. How much more votes did the second candidate get than the first candidate?
86,733 - 62,436 = 24,297 more votes for the second candidate.

## Review Exercise (book pages 20-21)

Choose the correct option and fill in the blanks.

1. (a) The sum of 36,529 and 41,372 is equal to:
(a) 77,904
(b) 77,903
(c) 77,901
(d) 77,902 ✓

1. (b) The sum of 17,278 and 62,354 is equal to:
(a) 78,234
(b) 342,211
(c) 79,632
(d) 213,455 ✓

1. (c) Ayesha had Rs 23,456. Her friend gave her Rs 13,131 more. Now, she has Rs _____.
(a) 36,587
(b) 35,467 ✓
(c) 36,434
(d) 34,567

1. (d) When we subtract 73,810 from 89,654 then we will get _____.
(a) 12,345
(b) 13,245

(c) 14,765
(d) 15,844 ✓

1. (e) In a pond, there were 87,654 fish. If 34,567 fish are shifted to another pond then _____ fish will be left in the first pond.
(a) 53,123
(b) 53,456
(c) 53,087 ✓
(d) 53,567

2. (a) 67,436 + 54,839
Ones place: 6 + 9 = 15. Write 5 and carry 1.

Tens place: 3 + 3 + 1 (carry) = 7. Write 7.

Hundreds place: 4 + 8 = 12. Write 2 and carry 1.

Thousands place: 7 + 4 + 1 (carry) = 12. Write 2 and carry 1.

Ten-thousands place: 6 + 5 + 1 (carry) = 12. Write 2 and carry 1.

Ten-thousands carry: write 1.

**Answer: 122,275.**

2. (b) 63,563 + 42,827
Ones place: 3 + 7 = 10. Write 0 and carry 1.

Tens place: 6 + 2 + 1 (carry) = 9. Write 9.

Hundreds place: 5 + 8 = 13. Write 3 and carry 1.

Thousands place: 3 + 2 + 1 (carry) = 6. Write 6.

Ten-thousands place: 6 + 4 = 10. Write 0 and carry 1.

Ten-thousands carry: write 1.

**Answer: 106,390.**

2. (c) 78,938 + 12,475
Ones place: 8 + 5 = 13. Write 3 and carry 1.

Tens place: 3 + 7 + 1 (carry) = 11. Write 1 and carry 1.

Hundreds place: 9 + 4 + 1 (carry) = 14. Write 4 and carry 1.

Thousands place: 8 + 2 + 1 (carry) = 11. Write 1 and carry 1.

Ten-thousands place: 7 + 1 + 1 (carry) = 9. Write 9.

**Answer: 91,413.**

2. (d) 67,343 - 41,235

Ones place: 3 is less than 5, so borrow 1 from the next place: 13 - 5 = 8. Write 8.

Tens place: 4 - 1 (borrow) = 3, then 3 - 3 = 0. Write 0.

Hundreds place: 3 - 2 = 1. Write 1.

Thousands place: 7 - 1 = 6. Write 6.

Ten-thousands place: 6 - 4 = 2. Write 2.

**Answer: 26,108.**

2. (e) 83,893 - 23,101

Ones place: 3 - 1 = 2. Write 2.

Tens place: 9 - 0 = 9. Write 9.

Hundreds place: 8 - 1 = 7. Write 7.

Thousands place: 3 - 3 = 0. Write 0.

Ten-thousands place: 8 - 2 = 6. Write 6.

**Answer: 60,792.**

2. (f) 33,275 - 26,238

Ones place: 5 is less than 8, so borrow 1 from the next place: 15 - 8 = 7. Write 7.

Tens place: 7 - 1 (borrow) = 6, then 6 - 3 = 3. Write 3.

Hundreds place: 2 - 2 = 0. Write 0.

Thousands place: 3 is less than 6, so borrow 1 from the next place: 13 - 6 = 7. Write 7.

Ten-thousands place: 3 - 1 (borrow) = 2, then 2 - 2 = 0. Write 0.

**Answer: 7,037.**

3. (a) 45,234 + 12,345
Ones place: 4 + 5 = 9. Write 9.

Tens place: 3 + 4 = 7. Write 7.

Hundreds place: 2 + 3 = 5. Write 5.

Thousands place: 5 + 2 = 7. Write 7.

Ten-thousands place: 4 + 1 = 5. Write 5.

**Answer: 57,579.**

3. (b) 24,567 + 13,466
Ones place: 7 + 6 = 13. Write 3 and carry 1.

Tens place: 6 + 6 + 1 (carry) = 13. Write 3 and carry 1.

Hundreds place: 5 + 4 + 1 (carry) = 10. Write 0 and carry 1.

Thousands place: 4 + 3 + 1 (carry) = 8. Write 8.

Ten-thousands place: 2 + 1 = 3. Write 3.

**Answer: 38,033.**

3. (c) 90,766 + 38,967
Ones place: 6 + 7 = 13. Write 3 and carry 1.

Tens place: 6 + 6 + 1 (carry) = 13. Write 3 and carry 1.

Hundreds place: 7 + 9 + 1 (carry) = 17. Write 7 and carry 1.

Thousands place: 0 + 8 + 1 (carry) = 9. Write 9.

Ten-thousands place: 9 + 3 = 12. Write 2 and carry 1.

Ten-thousands carry: write 1.

**Answer: 129,733.**

3. (d) 46,525 - 23,145
Ones place: 5 - 5 = 0. Write 0.

Tens place: 2 is less than 4, so borrow 1 from the next place: 12 - 4 = 8. Write 8.

Hundreds place: 5 - 1 (borrow) = 4, then 4 - 1 = 3. Write 3.

Thousands place: 6 - 3 = 3. Write 3.

Ten-thousands place: 4 - 2 = 2. Write 2.

**Answer: 23,380.**

3. (e) 76,247 - 74,166
Ones place: 7 - 6 = 1. Write 1.

Tens place: 4 is less than 6, so borrow 1 from the next place: 14 - 6 = 8. Write 8.

Hundreds place: 2 - 1 (borrow) = 1, then 1 - 1 = 0. Write 0.

Thousands place: 6 - 4 = 2. Write 2.

Ten-thousands place: 7 - 7 = 0. Write 0.

**Answer: 2,081.**

3. (f) 46,016 - 20,989
Ones place: 6 is less than 9, so borrow 1 from the next place: 16 - 9 = 7. Write 7.

Tens place: 1 - 1 (borrow) = 0, which is less than 8, so borrow 1 from the next place: 10 - 8 = 2. Write 2.

Hundreds place: 0 - 1 (borrow) = -1, which is less than 9, so borrow 1 from the next place: 9 - 9 = 0. Write 0.

Thousands place: 6 - 1 (borrow) = 5, then 5 - 0 = 5. Write 5.

Ten-thousands place: 4 - 2 = 2. Write 2.

**Answer: 25,027.**

4. (a) In the first week, 23,456 people went to visit the beach and in the second week 34,567 people went to visit the beach. Find the total number of people visited the beach in two weeks.
23,456 + 34,567 = 58,023 people.

4. (b) In which week less people visited the beach and by how much difference?
First week: 34,567 - 23,456 = 11,111 fewer people visited in the first week.

5. (a) There were 12,345 cattle in a farm. If 34,567 more cattle are added, then find how many cattle were there in the farm altogether?
12,345 + 34,567 = 46,912 cattle.

5. (b) If 26,754 were goats out of the total, then what was the number of cattle other than goats?
46,912 - 26,754 = 20,158 cattle other than goats.

6. There are 45,765 trees in a forest. If 32,124 are cactus trees, then find the number of trees other than cactus?
45,765 - 32,124 = 13,641 trees other than cactus.

7. Arsalan has Rs 51,346. He wants to buy a laptop which costs Rs 75,432. How much more amount does he need to buy the laptop?
75,432 - 51,346 = Rs 24,086 more needed.

## Unit 1 - Multiplication and Division (Section 3)

### Exercise 1 (book page 25)

Method: multiply by place value — multiply the multiplicand by each digit of the multiplier, then add the partial products.

1. (a) 631 x 4
Multiply 631 by 4 (4 ones): 631 x 4 = 2,524.

**Answer: 2,524.**

1. (b) 431 x 35
Multiply 431 by 5 (5 ones): 431 x 5 = 2,155.

Multiply 431 by 30 (3 tens): 431 x 30 = 12,930.

Add the partial products: 2,155 + 12,930 = 15,085.

**Answer: 15,085.**

1. (c) 8,434 x 31
Multiply 8,434 by 1 (1 ones): 8,434 x 1 = 8,434.

Multiply 8,434 by 30 (3 tens): 8,434 x 30 = 253,020.

Add the partial products: 8,434 + 253,020 = 261,454.

**Answer: 261,454.**

1. (d) 8,046 x 678
Multiply 8,046 by 8 (8 ones): 8,046 x 8 = 64,368.

Multiply 8,046 by 70 (7 tens): 8,046 x 70 = 563,220.

Multiply 8,046 by 600 (6 hundreds): 8,046 x 600 = 4,827,600.

Add the partial products: 64,368 + 563,220 + 4,827,600 = 5,455,188.

**Answer: 5,455,188.**

1. (e) 7,601 x 546
Multiply 7,601 by 6 (6 ones): 7,601 x 6 = 45,606.

Multiply 7,601 by 40 (4 tens): 7,601 x 40 = 304,040.

Multiply 7,601 by 500 (5 hundreds): 7,601 x 500 = 3,800,500.

Add the partial products: 45,606 + 304,040 + 3,800,500 = 4,150,146.

**Answer: 4,150,146.**

1. (f) 41,175 x 80
Multiply 41,175 by 80 (8 tens): 41,175 x 80 = 3,294,000.

**Answer: 3,294,000.**

1. (g) 79,762 x 15
Multiply 79,762 by 5 (5 ones): 79,762 x 5 = 398,810.

Multiply 79,762 by 10 (1 tens): 79,762 x 10 = 797,620.

Add the partial products: 398,810 + 797,620 = 1,196,430.

**Answer: 1,196,430.**

1. (h) 63,506 x 303
Multiply 63,506 by 3 (3 ones): 63,506 x 3 = 190,518.

Multiply 63,506 by 300 (3 hundreds): 63,506 x 300 = 19,051,800.

Add the partial products: 190,518 + 19,051,800 = 19,242,318.

**Answer: 19,242,318.**

1. (i) 11,098 x 237
Multiply 11,098 by 7 (7 ones): 11,098 x 7 = 77,686.

Multiply 11,098 by 30 (3 tens): 11,098 x 30 = 332,940.

Multiply 11,098 by 200 (2 hundreds): 11,098 x 200 = 2,219,600.

Add the partial products: 77,686 + 332,940 + 2,219,600 = 2,630,226.

**Answer: 2,630,226.**

2. A shopkeeper sold 34,523 m cloth in a week. How much cloth will he sell in 21 weeks?
34,523 x 21 = 34,523 x 1 + 34,523 x 20 = 34,523 + 690,460 = 724,983 m.

3. (a) Majid earns Rs 11,045 in a day. How much money will he earn in 365 days?
11,045 x 365 = Rs 4,031,425.

3. (b) How much money will he earn in 2 years?
2 years = 730 days. 11,045 x 730 = Rs 8,062,850.

4. In a factory, 20,134 notebooks were printed in a day. How many notebooks will be printed in 210 days?
20,134 x 210 = 4,228,140 notebooks.

5. Each member of a group give Rs 34,156 for a tour of Naran and Kagan. If there are 345 members of the group, then how much money will the group

collect altogether?

34,156 x 345 = Rs 11,783,820.

Try It! (book page 29): the numbers in the boxes are the products of the numbers in the two circles next to them. 2,808 = 234 x 12, so the top circle is 234; then 2,340 = 234 x 10, so the other blank circle is 10.

## Exercise 2 (book page 29)

Method: long division — divide, multiply, subtract, bring down, repeat; the last remainder is written with the quotient.

1. (a) 585 / 3

3 into 5 goes 1 time (1 x 3 = 3), remainder 2.

3 into 28 goes 9 times (9 x 3 = 27), remainder 1.

3 into 15 goes 5 times (5 x 3 = 15), remainder 0.

**Answer: Quotient = 195, Remainder = 0.**

1. (b) 1,816 / 4

4 into 18 goes 4 times (4 x 4 = 16), remainder 2.

4 into 21 goes 5 times (5 x 4 = 20), remainder 1.

4 into 16 goes 4 times (4 x 4 = 16), remainder 0.

**Answer: Quotient = 454, Remainder = 0.**

1. (c) 6,972 / 42

42 into 69 goes 1 time (1 x 42 = 42), remainder 27.

42 into 277 goes 6 times (6 x 42 = 252), remainder 25.

42 into 252 goes 6 times (6 x 42 = 252), remainder 0.

**Answer: Quotient = 166, Remainder = 0.**

1. (d) 7,546 / 22

22 into 75 goes 3 times (3 x 22 = 66), remainder 9.

22 into 94 goes 4 times (4 x 22 = 88), remainder 6.

22 into 66 goes 3 times (3 x 22 = 66), remainder 0.

**Answer: Quotient = 343, Remainder = 0.**

1. (e) 9,568 / 23

23 into 95 goes 4 times (4 x 23 = 92), remainder 3.

23 into 36 goes 1 time (1 x 23 = 23), remainder 13.

23 into 138 goes 6 times (6 x 23 = 138), remainder 0.

**Answer: Quotient = 416, Remainder = 0.**

1. (f) 9,641 / 31

31 into 96 goes 3 times (3 x 31 = 93), remainder 3.

31 into 34 goes 1 time (1 x 31 = 31), remainder 3.

31 into 31 goes 1 time (1 x 31 = 31), remainder 0.

**Answer: Quotient = 311, Remainder = 0.**

1. (g) 2,868 / 12

12 into 28 goes 2 times (2 x 12 = 24), remainder 4.

12 into 46 goes 3 times (3 x 12 = 36), remainder 10.

12 into 108 goes 9 times (9 x 12 = 108), remainder 0.

**Answer: Quotient = 239, Remainder = 0.**

1. (h) 7,392 / 32

32 into 73 goes 2 times (2 x 32 = 64), remainder 9.

32 into 99 goes 3 times (3 x 32 = 96), remainder 3.

32 into 32 goes 1 time (1 x 32 = 32), remainder 0.

**Answer: Quotient = 231, Remainder = 0.**

1. (i) 133 / 11

11 into 13 goes 1 time (1 x 11 = 11), remainder 2.

11 into 23 goes 2 times (2 x 11 = 22), remainder 1.

**Answer: Quotient = 12, Remainder = 1.**

1. (j) 1,056 / 8
8 into 10 goes 1 time (1 x 8 = 8), remainder 2.

8 into 25 goes 3 times (3 x 8 = 24), remainder 1.

8 into 16 goes 2 times (2 x 8 = 16), remainder 0.

**Answer: Quotient = 132, Remainder = 0.**

1. (k) 1,848 / 88
88 into 184 goes 2 times (2 x 88 = 176), remainder 8.

88 into 88 goes 1 time (1 x 88 = 88), remainder 0.

**Answer: Quotient = 21, Remainder = 0.**

1. (l) 4,662 / 42
42 into 46 goes 1 time (1 x 42 = 42), remainder 4.

42 into 46 goes 1 time (1 x 42 = 42), remainder 4.

42 into 42 goes 1 time (1 x 42 = 42), remainder 0.

**Answer: Quotient = 111, Remainder = 0.**

1. (m) 6,125 / 10
10 into 61 goes 6 times (6 x 10 = 60), remainder 1.

10 into 12 goes 1 time (1 x 10 = 10), remainder 2.

10 into 25 goes 2 times (2 x 10 = 20), remainder 5.

**Answer: Quotient = 612, Remainder = 5.**

1. (n) 2,060 / 23
23 into 206 goes 8 times (8 x 23 = 184), remainder 22.

23 into 220 goes 9 times (9 x 23 = 207), remainder 13.

**Answer: Quotient = 89, Remainder = 13.**

2. In 45 relief camps, 2,205 blankets were distributed. How many blankets did each camp get?
2,205 / 45 = 49 blankets per camp.

3. If 1,107 chairs are placed in 27 rows, then how many chairs will be there in a row?
1,107 / 27 = 41 chairs in a row.

4. If 3,036 biscuits are packed in 11 boxes, then find out how many biscuits are there in a box?
3,036 / 11 = 276 biscuits in a box.

5. If 6,666 books are to be kept in 33 cupboards in a library, then how many books will be there in each cupboard?
6,666 / 33 = 202 books in each cupboard.

6. (a) Saad bought 10 washing machines for Rs 78,950 and an oven for Rs 21,550. How much money did he spend altogether?
78,950 + 21,550 = Rs 100,500.

6. (b) How much more amount did he spend on washing machines than an oven?
78,950 - 21,550 = Rs 57,400 more on washing machines.

6. (c) How much amount did he spend on a washing machine?
78,950 / 10 = Rs 7,895 per washing machine.

7. (a) In 30 bags, 1,350 kg rice are packed. How many kilograms of rice are in one bag?
1,350 / 30 = 45 kg in one bag.

7. (b) How many kilograms of rice will be packed in 38 bags?
45 x 38 = 1,710 kg in 38 bags.

## Exercise 3 - Patterns (book page 32)

1. (a) 11, 15, 19, 23, 27, __, __
Rule: adding 4 each time. Next two terms: 31, 35.

1. (b) 30, 60, 90, 120, 150, __, __
Rule: adding 30 each time. Next two terms: 180, 210.

1. (c) 6, 12, 18, 24, 30, __, __
Rule: adding 6 each time. Next two terms: 36, 42.

1. (d) 850, 800, 750, 700, 650, __, __
Rule: subtracting 50 each time. Next two terms: 600, 550.

1. (e) 106, 103, 100, 97, 94, __, __
Rule: subtracting 3 each time. Next two terms: 91, 88.

1. (f) 284, 288, 292, 296, __, __
Rule: adding 4 each time. Next two terms: 300, 304.

1. (g) 560, 540, 520, 500, __, __
Rule: subtracting 20 each time. Next two terms: 480, 460.

2. Observe the given hundreds chart and find at least 5 patterns. Also set the rules for these patterns.
Five patterns in the chart: (1) Moving right across a row adds 1 each time (for example 1, 2, 3, 4). (2) Moving down a column adds 10 each time (for example 1, 11, 21, 31). (3) Moving left across a row subtracts 1 each time. (4) Moving up a column subtracts 10 each time. (5) Moving diagonally down-right adds 11 each time (for example 1, 12, 23, 34). Other correct patterns: even numbers 2, 4, 6, 8 and multiples of 10: 10, 20, 30.

3. (a) Observe the table: Weeks 1, 2, 3, 4, 5 and Height of the plant 4 cm, 8 cm, 12 cm, 16 cm, 20 cm. Describe the rule of pattern.
Rule: multiply the number of weeks by 4. Height = 4 x weeks (1 x 4 = 4 cm, 2 x 4 = 8 cm, 3 x 4 = 12 cm, 4 x 4 = 16 cm, 5 x 4 = 20 cm).

3. (b) Observe the table: Boxes of blocks 1, 2, 3, 4, 5 and Total number of blocks 20, 40, 60, 80, 100. Describe the rule of pattern.
Rule: multiply the number of boxes by 20. Total blocks = 20 x boxes (1 x 20 = 20, 2 x 20 = 40, 3 x 20 = 60, 4 x 20 = 80, 5 x 20 = 100).

## Review Exercise (book pages 33-34)

1. (a) There are 4,500 plants in 90 rows. Each row contains equal number of plants. Find the number of plants in a row.
(a) 10
(b) 5
(c) 100
(d) 50 ✓

1. (b) If the price of one book is Rs 250, then the price of 22 books will be:
(a) Rs 5,550
(b) Rs 5,555
(c) Rs 5,500 ✓
(d) Rs 5,000

1. (c) By dividing 3,960 by 88, we will get:
(a) 47

(b) 46
(c) 45
(d) 41

1. (d) The next term in 6, 18, 30, 42, is:
(a) 48
(b) 54
(c) 56
(d) 46

1. (e) The next term in 88, 78, 68, is:
(a) 58
(b) 47
(c) 48
(d) 98

2. (a) 245 x 2
Multiply 245 by 2 (2 ones): 245 x 2 = 490.

**Answer: 490.**

2. (b) 743 x 12
Multiply 743 by 2 (2 ones): 743 x 2 = 1,486.

Multiply 743 by 10 (1 tens): 743 x 10 = 7,430.

Add the partial products: 1,486 + 7,430 = 8,916.

**Answer: 8,916.**

2. (c) 4,324 x 41
Multiply 4,324 by 1 (1 ones): 4,324 x 1 = 4,324.

Multiply 4,324 by 40 (4 tens): 4,324 x 40 = 172,960.

Add the partial products: 4,324 + 172,960 = 177,284.

**Answer: 177,284.**

2. (d) 1,245 x 13
Multiply 1,245 by 3 (3 ones): 1,245 x 3 = 3,735.

Multiply 1,245 by 10 (1 tens): 1,245 x 10 = 12,450.

Add the partial products: 3,735 + 12,450 = 16,185.

**Answer: 16,185.**

2. (e) 67,453 x 345
Multiply 67,453 by 5 (5 ones): 67,453 x 5 = 337,265.

Multiply 67,453 by 40 (4 tens): 67,453 x 40 = 2,698,120.

Multiply 67,453 by 300 (3 hundreds): 67,453 x 300 = 20,235,900.

Add the partial products: 337,265 + 2,698,120 + 20,235,900 = 23,271,285.

**Answer: 23,271,285.**

2. (f) 78,965 x 453
Multiply 78,965 by 3 (3 ones): 78,965 x 3 = 236,895.

Multiply 78,965 by 50 (5 tens): 78,965 x 50 = 3,948,250.

Multiply 78,965 by 400 (4 hundreds): 78,965 x 400 = 31,586,000.

Add the partial products: 236,895 + 3,948,250 + 31,586,000 = 35,771,145.

**Answer: 35,771,145.**

3. (a) 380 / 5
5 into 38 goes 7 times (7 x 5 = 35), remainder 3.

5 into 30 goes 6 times (6 x 5 = 30), remainder 0.

**Answer: Quotient = 76, Remainder = 0.**

3. (b) 196 / 12
12 into 19 goes 1 time (1 x 12 = 12), remainder 7.

12 into 76 goes 6 times (6 x 12 = 72), remainder 4.

**Answer: Quotient = 16, Remainder = 4.**

3. (c) 2,925 / 6
6 into 29 goes 4 times (4 x 6 = 24), remainder 5.

6 into 52 goes 8 times (8 x 6 = 48), remainder 4.

6 into 45 goes 7 times (7 x 6 = 42), remainder 3.

**Answer: Quotient = 487, Remainder = 3.**

3. (d) 3,294 / 61
61 into 329 goes 5 times (5 x 61 = 305), remainder 24.

61 into 244 goes 4 times (4 x 61 = 244), remainder 0.

**Answer: Quotient = 54, Remainder = 0.**

3. (e) 1,766 / 22
22 into 176 goes 8 times (8 x 22 = 176), remainder 0.

Bring down 6: we have 6, which is less than 22, so write 0 in the quotient.

**Answer: Quotient = 80, Remainder = 6.**

3. (f) 2,205 / 49
49 into 220 goes 4 times (4 x 49 = 196), remainder 24.

49 into 245 goes 5 times (5 x 49 = 245), remainder 0.

**Answer: Quotient = 45, Remainder = 0.**

4. A bus has the capacity of 45 passengers. How many buses would be needed for 1,575 passengers?
1,575 / 45 = 35 buses.

5. (a) A car covers a distance of 1,288 km in 23 hours. How much distance would it cover in one hour?
1,288 / 23 = 56 km in one hour.

5. (b) How much distance would it cover in 11 hours?
56 x 11 = 616 km in 11 hours.

6. (a) A man pays Rs 23,452 as one month installment of the car. How much amount will he pay in 2 years?
2 years = 24 months. 23,452 x 24 = Rs 562,848.

6. (b) How much amount will he pay in 3 years?
3 years = 36 months. 23,452 x 36 = Rs 844,272.

7. Zaeem has 1,867 lego blocks. His sister gives him 4 more boxes of lego blocks. There are 1,205 lego blocks in each box. How many lego blocks Zaeem has in total?
4 boxes give 4 x 1,205 = 4,820 blocks. Total = 1,867 + 4,820 = 6,687 blocks.

8. (a) 3, 9, 15, 21, __, __
Rule: adding 6 each time. Next two terms: 27, 33.

8. (b) 100, 90, 80, 70, 60, __, __
Rule: subtracting 10 each time. Next two terms: 50, 40.

8. (c) 12, 18, 24, 30, 36, __, __
Rule: adding 6 each time. Next two terms: 42, 48.

8. (d) 2, 10, 18, 26, 34, __, __
Rule: adding 8 each time. Next two terms: 42, 50.

8. (e) 106, 95, 84, 73, 62, __, __
Rule: subtracting 11 each time. Next two terms: 51, 40.

## Unit 2 - Factors and Multiples

### Exercise 1 (book page 38)

1. Tick the numbers that are divisible by 2: (a) 16, (b) 43, (c) 98, (d) 134, (e) 6,781, (f) 2,114, (g) 9,226, (h) 67, (i) 540, (j) 82,420.
Divisible by 2 (even numbers): 16, 98, 134, 2,114, 9,226, 540, 82,420. Not divisible by 2: 43, 6,781, 67.

2. Tick the numbers that are divisible by 3: (a) 20, (b) 27, (c) 165, (d) 125, (e) 6,720, (f) 8,955, (g) 52,110, (h) 21,945.
Divisible by 3 (digit sum divisible by 3): 27 (2+7=9), 165 (1+6+5=12), 6,720 (6+7+2+0=15), 8,955 (8+9+5+5=27), 52,110 (5+2+1+1+0=9), 21,945 (2+1+9+4+5=21). Not divisible by 3: 20 (2), 125 (8).

3. Tick the numbers that are divisible by 5: (a) 30, (b) 50, (c) 7,895, (d) 2,298.
Divisible by 5 (end in 0 or 5): 30, 50, 7,895. Not divisible by 5: 2,298.

4. Tick the numbers that are divisible by 10: (a) 56,560, (b) 1,982, (c) 42,420, (d) 130.
Divisible by 10 (end in 0): 56,560, 42,420, 130. Not divisible by 10: 1,982.

## Exercise 2 (book pages 41-42)

1. Write all composite numbers between 30 and 50.

Composite numbers between 30 and 50: 32, 33, 34, 35, 36, 38, 39, 40, 42, 44, 45, 46, 48, 49.

2. Encircle the prime numbers: (a) 15, (b) 31, (c) 42, (d) 67, (e) 11, (f) 52, (g) 98, (h) 89.

Prime numbers: 31, 67, 11, 89. (15, 42, 52, 98 are composite.)

3. Write the first 15 prime numbers.

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47.

4. Identify the composite numbers and colour them (grid of numbers from 29 to 100).

Colour every number except the prime numbers 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97 — all the other numbers in the grid are composite.

5. Find the factors of 15.

Factors of 15: 1, 3, 5, 15.

5. Find the factors of 32.

Factors of 32: 1, 2, 4, 8, 16, 32.

5. Find the factors of 27.

Factors of 27: 1, 3, 9, 27.

5. Find the factors of 12.

Factors of 12: 1, 2, 3, 4, 6, 12.

5. Find the factors of 10.

Factors of 10: 1, 2, 5, 10.

5. Find the factors of 38.

Factors of 38: 1, 2, 19, 38.

5. Find the factors of 22.

Factors of 22: 1, 2, 11, 22.

5. Find the factors of 49.

Factors of 49: 1, 7, 49.

5. Find the factors of 40.

Factors of 40: 1, 2, 4, 5, 8, 10, 20, 40.

5. Find the factors of 6.
Factors of 6: 1, 2, 3, 6.

6. Find the first 10 multiples of 8.
First 10 multiples of 8: 8, 16, 24, 32, 40, 48, 56, 64, 72, 80.

6. Find the first 10 multiples of 2.
First 10 multiples of 2: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20.

6. Find the first 10 multiples of 3.
First 10 multiples of 3: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30.

6. Find the first 10 multiples of 5.
First 10 multiples of 5: 5, 10, 15, 20, 25, 30, 35, 40, 45, 50.

6. Find the first 10 multiples of 7.
First 10 multiples of 7: 7, 14, 21, 28, 35, 42, 49, 56, 63, 70.

6. Find the first 10 multiples of 6.
First 10 multiples of 6: 6, 12, 18, 24, 30, 36, 42, 48, 54, 60.

6. Find the first 10 multiples of 4.
First 10 multiples of 4: 4, 8, 12, 16, 20, 24, 28, 32, 36, 40.

6. Find the first 10 multiples of 9.
First 10 multiples of 9: 9, 18, 27, 36, 45, 54, 63, 72, 81, 90.

## Exercise 3 (book pages 45-46)

1. Find the prime factors of 17.
Prime factorization of 17 = 17 (17 is itself a prime number).

1. Find the prime factors of 21.
Prime factorization of 21 = 3 x 7.

1. Find the prime factors of 18.
Prime factorization of 18 = 2 x 3 x 3.

1. Find the prime factors of 34.
Prime factorization of 34 = 2 x 17.

1. Find the prime factors of 44.
Prime factorization of 44 = 2 x 2 x 11.

1. Find the prime factors of 33.
Prime factorization of 33 = 3 x 11.

1. Find the prime factors of 4.
Prime factorization of 4 = 2 x 2.

1. Find the prime factors of 39.
Prime factorization of 39 = 3 x 13.

1. Find the prime factors of 14.
Prime factorization of 14 = 2 x 7.

1. Find the prime factors of 48.
Prime factorization of 48 = 2 x 2 x 2 x 2 x 3.

2. Find the common prime factors of 10, 20.
Prime factorization: 10 = 2 x 5; 20 = 2 x 2 x 5. Common prime factors: 2, 5.

2. Find the common prime factors of 6, 18.
Prime factorization: 6 = 2 x 3; 18 = 2 x 3 x 3. Common prime factors: 2, 3.

2. Find the common prime factors of 24, 32, 18.
Prime factorization: 24 = 2 x 2 x 2 x 3; 32 = 2 x 2 x 2 x 2 x 2; 18 = 2 x 3 x 3. Common prime factors: 2.

2. Find the common prime factors of 14, 30.
Prime factorization: 14 = 2 x 7; 30 = 2 x 3 x 5. Common prime factors: 2.

2. Find the common prime factors of 7, 21, 28.
Prime factorization: 7 = 7; 21 = 3 x 7; 28 = 2 x 2 x 7. Common prime factors: 7.

2. Find the common prime factors of 20, 25, 15.
Prime factorization: 20 = 2 x 2 x 5; 25 = 5 x 5; 15 = 3 x 5. Common prime factors: 5.

2. Find the common prime factors of 13, 39.
Prime factorization: 13 = 13; 39 = 3 x 13. Common prime factors: 13.

2. Find the common prime factors of 5, 30, 12.
Prime factorization: 5 = 5; 30 = 2 x 3 x 5; 12 = 2 x 2 x 3. Common prime factors: none — no prime factor is common to all of them.

2. Find the common prime factors of 4, 8.
Prime factorization: 4 = 2 x 2; 8 = 2 x 2 x 2. Common prime factors: 2, 2.

3. Find the first common multiple of 10, 20, 30.
The first common multiple of 10, 20, 30 is 60.

3. Find the first common multiple of 3, 5.
The first common multiple of 3, 5 is 15.

3. Find the first common multiple of 9, 12.
The first common multiple of 9, 12 is 36.

3. Find the first common multiple of 51, 17, 34.
The first common multiple of 51, 17, 34 is 102.

3. Find the first common multiple of 12, 22.
The first common multiple of 12, 22 is 132.

3. Find the first common multiple of 8, 4, 16.
The first common multiple of 8, 4, 16 is 16.

3. Find the first common multiple of 7, 14.
The first common multiple of 7, 14 is 14.

3. Find the first common multiple of 6, 15.
The first common multiple of 6, 15 is 30.

3. Find the first common multiple of 2, 5, 10.
The first common multiple of 2, 5, 10 is 10.

## Review Exercise (book pages 47-48)

1. (a) 13 is a _____ number.
(a) composite
(b) multiple
(c) common
(d) prime ✓

1. (b) If _____ of all the digits of a number is divisible by 3, then that number is divisible by 3.
(a) sum ✓
(b) product
(c) difference
(d) quotient

1. (c) Prime factorization of 24 is:

(a) 1 x 24

(b) 8 x 3

(c) 2 x 2 x 2 x 3

(d) 2 x 6 x 2

1. (d) The common prime factor of 2 and 4 is:

(a) 2

(b) 8

(c) 1

(d) 4

1. (e) The first common multiple of 5 and 10 is:

(a) 20

(b) 5

(c) 10

(d) 50

2. Tick the given boxes in the table by using divisibility rule. Numbers: 112, 986, 5,409, 5,600, 81,810, 5,912, 53,800, 2,134.

112: divisible by 2 986: divisible by 2 5,409: divisible by 3 5,600: divisible by 2, 5, 10 81,810: divisible by 2, 3, 5, 10 5,912: divisible by 2 53,800: divisible by 2, 5, 10 2,134: divisible by 2.

3. Write the first 12 composite numbers.

4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20, 21.

4. Write prime numbers between 21 and 60.

23, 29, 31, 37, 41, 43, 47, 53, 59.

5. Find the factors of 10.

Factors of 10: 1, 2, 5, 10.

5. Find the factors of 25.

Factors of 25: 1, 5, 25.

5. Find the factors of 35.

Factors of 35: 1, 5, 7, 35.

5. Find the factors of 46.

Factors of 46: 1, 2, 23, 46.

5. Find the factors of 23.

Factors of 23: 1, 23.

5. Find the factors of 16.
Factors of 16: 1, 2, 4, 8, 16.

5. Find the factors of 4.
Factors of 4: 1, 2, 4.

5. Find the factors of 47.
Factors of 47: 1, 47.

5. Find the factors of 38.
Factors of 38: 1, 2, 19, 38.

5. Find the factors of 20.
Factors of 20: 1, 2, 4, 5, 10, 20.

6. Find the first 6 multiples of 2.
First 6 multiples of 2: 2, 4, 6, 8, 10, 12.

6. Find the first 6 multiples of 6.
First 6 multiples of 6: 6, 12, 18, 24, 30, 36.

6. Find the first 6 multiples of 5.
First 6 multiples of 5: 5, 10, 15, 20, 25, 30.

6. Find the first 6 multiples of 9.
First 6 multiples of 9: 9, 18, 27, 36, 45, 54.

7. Find the prime factors of 5.
Prime factorization of 5 = 5.

7. Find the prime factors of 19.
Prime factorization of 19 = 19.

7. Find the prime factors of 22.
Prime factorization of 22 = 2 x 11.

7. Find the prime factors of 15.
Prime factorization of 15 = 3 x 5.

7. Find the prime factors of 21.
Prime factorization of 21 = 3 x 7.

7. Find the prime factors of 40.

Prime factorization of 40 = 2 x 2 x 2 x 5.

7. Find the prime factors of 30.
Prime factorization of 30 = 2 x 3 x 5.

7. Find the prime factors of 8.
Prime factorization of 8 = 2 x 2 x 2.

7. Find the prime factors of 41.
Prime factorization of 41 = 41.

7. Find the prime factors of 38.
Prime factorization of 38 = 2 x 19.

8. Find the common prime factors of 4, 20.
Common prime factors: 2, 2.

8. Find the common prime factors of 16, 24.
Common prime factors: 2, 2, 2.

8. Find the common prime factors of 28, 56, 14.
Common prime factors: 2, 7.

8. Find the common prime factors of 17, 34.
Common prime factors: 17.

8. Find the common prime factors of 12, 6, 18.
Common prime factors: 2, 3.

8. Find the common prime factors of 5, 10, 20.
Common prime factors: 5.

9. Find the first common multiple of 2, 7.
The first common multiple of 2, 7 is 14.

9. Find the first common multiple of 6, 10.
The first common multiple of 6, 10 is 30.

9. Find the first common multiple of 12, 14, 18.
The first common multiple of 12, 14, 18 is 252.

9. Find the first common multiple of 15, 30.
The first common multiple of 15, 30 is 30.

9. Find the first common multiple of 5, 15, 20.
The first common multiple of 5, 15, 20 is 60.

9. Find the first common multiple of 6, 12, 15.
The first common multiple of 6, 12, 15 is 60.

## Unit 3 - Fractions

### Exercise 1 (book pages 58-59)

1. Encircle the unlike fractions of the following: (a) 3/5, 1/2; (b) 7/9, 4/9; (c) 6/11, 1/11; (d) 2/8, 3/8; (e) 6/10, 1/5; (f) 5/9, 2/7.
Unlike fractions have different denominators. Unlike pairs: (a) 3/5 and 1/2 (denominators 5 and 2), (e) 6/10 and 1/5 (denominators 10 and 5), (f) 5/9 and 2/7 (denominators 9 and 7). Like pairs (same denominator): (b) 7/9 and 4/9, (c) 6/11 and 1/11, (d) 2/8 and 3/8.

2. Compare the given fractions and write symbols of <, > or =: (a) 1/2 __ 3/6; (b) 4/5 __ 9/10; (c) 6/12 __ 1/2; (d) 1/7 __ 5/8; (e) 2/9 __ 5/6; (f) 8/12 __ 5/7; (g) 1/3 __ 1/4; (h) 4/11 __ 7/10.
(a) 1/2 = 3/6. (b) 4/5 = 8/10 < 9/10. (c) 6/12 = 1/2. (d) 1/7 = 8/56 < 35/56 = 5/8. (e) 2/9 = 4/18 < 15/18 = 5/6. (f) 8/12 = 56/84 < 60/84 = 5/7. (g) 1/3 = 4/12 > 3/12 = 1/4. (h) 4/11 = 40/110 < 77/110 = 7/10.

3. Write the following fractions into the lowest form: (a) 4/20; (b) 2/12; (c) 30/45; (d) 9/27; (e) 12/16; (f) 15/25; (g) 16/24; (h) 4/18; (i) 14/20; (j) 17/34.
(a) 4/20 = 1/5; (b) 2/12 = 1/6; (c) 30/45 = 2/3; (d) 9/27 = 1/3; (e) 12/16 = 3/4; (f) 15/25 = 3/5; (g) 16/24 = 2/3; (h) 4/18 = 2/9; (i) 14/20 = 7/10; (j) 17/34 = 1/2.

4. Encircle the proper fractions among the given fractions and tick the mixed number: (a) 2/5; (b) 7/8; (c) 3/4; (d) 3 4/7; (e) 1 1/2; (f) 3/8; (g) 9/11; (h) 3 7/11; (i) 6/7; (j) 11 1/2.
Proper fractions (numerator smaller than denominator): 2/5, 7/8, 3/4, 3/8, 9/11, 6/7. Mixed numbers: 3 4/7, 1 1/2, 3 7/11, 11 1/2.

5. Convert improper fraction 8/5 into a mixed number.
8/5 = 1 3/5.

5. Convert improper fraction 11/5 into a mixed number.
11/5 = 2 1/5.

5. Convert improper fraction 13/10 into a mixed number.

13/10 = 1 3/10.

5. Convert improper fraction 20/9 into a mixed number.
20/9 = 2 2/9.

5. Convert improper fraction 15/2 into a mixed number.
15/2 = 7 1/2.

6. Convert mixed number 2 3/5 into an improper fraction.
2 3/5 = 13/5.

6. Convert mixed number 7 5/6 into an improper fraction.
7 5/6 = 47/6.

6. Convert mixed number 4 1/7 into an improper fraction.
4 1/7 = 29/7.

6. Convert mixed number 5 3/11 into an improper fraction.
5 3/11 = 58/11.

6. Convert mixed number 6 1/3 into an improper fraction.
6 1/3 = 19/3.

6. Convert mixed number 2 4/13 into an improper fraction.
2 4/13 = 30/13.

7. (a) Write 3/5, 3/9, 3/7 in ascending and descending order.
Same numerator: the larger the denominator, the smaller the fraction.
Ascending: 3/9, 3/7, 3/5. Descending: 3/5, 3/7, 3/9.

7. (b) Write 3/4, 1/3, 6/7 in ascending and descending order.
With denominator 84: 3/4 = 63/84, 1/3 = 28/84, 6/7 = 72/84. Ascending: 1/3, 3/4, 6/7. Descending: 6/7, 3/4, 1/3.

7. (c) Write 3/5, 2/10, 4/15 in ascending and descending order.
With denominator 30: 3/5 = 18/30, 2/10 = 6/30, 4/15 = 8/30. Ascending: 2/10, 4/15, 3/5. Descending: 3/5, 4/15, 2/10.

8. Ali has three full and one half pizza. How can we write this in mixed number?
Three full and one half pizza = 3 1/2.

9. Mahad buys 1 1/2 kg of mangoes. Write this in improper fraction.
1 1/2 = (1 x 2 + 1)/2 = 3/2 kg.

Try It! (book page 58): I am a mixed number between 2 and 10. I am nearer to 8 than 4. If you separate my fractional part then I am an odd number. Who am I? Model answer: 7 1/2 (whole part 7 is odd, and 7 1/2 is nearer to 8 than to 4).

## Exercise 2 (book page 62)

1. (a) 6/7 + 5/7 (lowest form).
6/7 + 5/7 = 11/7 = 1 4/7.

1. (b) 11/13 + 11/13 (lowest form).
11/13 + 11/13 = 22/13 = 1 9/13.

1. (c) 5/17 + 11/17 (lowest form).
5/17 + 11/17 = 16/17 = 16/17.

1. (d) 7/15 + 8/15 (lowest form).
7/15 + 8/15 = 15/15 = 1.

1. (e) 5/16 + 5/16 (lowest form).
5/16 + 5/16 = 10/16 = 5/8.

1. (f) 2/19 + 12/19 (lowest form).
2/19 + 12/19 = 14/19 = 14/19.

2. (a) Subtract the smallest fraction from the greatest fraction: 2/3, 3/3.
Greatest minus smallest = 1 - 2/3 = 1/3.

2. (b) Subtract the smallest fraction from the greatest fraction: 1/11, 7/11.
Greatest minus smallest = 7/11 - 1/11 = 6/11.

2. (c) Subtract the smallest fraction from the greatest fraction: 11/12, 7/12.
Greatest minus smallest = 11/12 - 7/12 = 1/3.

2. (d) Subtract the smallest fraction from the greatest fraction: 7/21, 15/21.
Greatest minus smallest = 5/7 - 1/3 = 8/21.

2. (e) Subtract the smallest fraction from the greatest fraction: 2/5, 4/5.
Greatest minus smallest = 4/5 - 2/5 = 2/5.

2. (f) Subtract the smallest fraction from the greatest fraction: 8/10, 4/10.
Greatest minus smallest = 4/5 - 2/5 = 2/5.

2. (g) Subtract the smallest fraction from the greatest fraction: 2/13, 1/13.
Greatest minus smallest = 2/13 - 1/13 = 1/13.

2. (h) Subtract the smallest fraction from the greatest fraction: 5/15, 3/15.
Greatest minus smallest = 1/3 - 1/5 = 2/15.

3. (a) A painter paints 7/13 part of the wall in the first day and 3/13 part on the second day. How much wall does he paint in two days?
7/13 + 3/13 = 10/13 part of the wall.

3. (b) On which day does he paint more and how much?
First day: 7/13 - 3/13 = 4/13 part more on the first day.

4. 5/14 kg of artificial fertilizer and 7/14 kg of natural fertilizer have been used in a field. How much quantity of both fertilizers is used?
5/14 + 7/14 = 12/14 = 6/7 kg.

5. Saba did her maths homework in 2/10 hours and her sister did her maths homework in 7/10 hours. How many hours did both take to complete their homework?
2/10 + 7/10 = 9/10 hours.

6. Shiraz and Omar invest money in a business. Shiraz gets 7/11 share and Omar gets 10/11 share of the profit. Who has more share and how much?
Omar has more: 10/11 - 7/11 = 3/11 share more.

## Exercise 3 (book page 66)

1. (a) Multiply: 6/7 x 4.
6/7 x 4 = 3 3/7.

1. (b) Multiply: 9 x 5/6.
9 x 5/6 = 7 1/2.

1. (c) Multiply: 13/11 x 11.
13/11 x 11 = 13.

1. (d) Multiply: 8/9 x 6.
8/9 x 6 = 5 1/3.

1. (e) Multiply: 1/10 x 2.
1/10 x 2 = 1/5.

1. (f) Multiply: 6/13 x 1.
6/13 x 1 = 6/13.

1. (g) Multiply: 7/11 x 2.

7/11 x 2 = 1 3/11.

1. (h) Multiply: 3 2/3 x 1.
3 2/3 x 1 = 3 2/3.

2. (a) Solve: 6/2 x 3/6.
6/2 x 3/6 = 1 1/2.

2. (b) Solve: 9/11 x 5/10.
9/11 x 5/10 = 9/22.

2. (c) Solve: 3/17 x 3 3/4.
3/17 x 3 3/4 = 45/68.

2. (d) Solve: 7 1/7 x 1 5/8.
7 1/7 x 1 5/8 = 11 17/28.

2. (e) Solve: 2/9 x 1 5/6 x 5/6.
2/9 x 1 5/6 x 5/6 = 55/162.

2. (f) Solve: 8/12 x 3 8/11 x 5/7.
8/12 x 3 8/11 x 5/7 = 1 179/231.

2. (g) Solve: 4/3 x 1/4 x 7 7/10.
4/3 x 1/4 x 7 7/10 = 2 17/30.

3. (a) Solve: 7/20 / 2.
7/20 / 2 = 7/40 (divide by 2 means multiply by 1/2).

3. (b) Solve: 2/15 / 5.
2/15 / 5 = 2/75 (divide by 5 means multiply by 1/5).

3. (c) Solve: 20/35 / 9.
20/35 / 9 = 4/63 (divide by 9 means multiply by 1/9).

3. (d) Solve: 21/27 / 3.
21/27 / 3 = 7/27 (divide by 3 means multiply by 1/3).

3. (e) Solve: 14/16 / 7.
14/16 / 7 = 1/8 (divide by 7 means multiply by 1/7).

3. (f) Solve: 15/20 / 21.
15/20 / 21 = 1/28 (divide by 21 means multiply by 1/21).

3. (g) Solve: 18/24 / 3.
18/24 / 3 = 1/4 (divide by 3 means multiply by 1/3).

3. (h) Solve: 14/18 / 18.
14/18 / 18 = 7/162 (divide by 18 means multiply by 1/18).

4. If the weight of 5 packets of sugar is 4 7/8 kg, then what will be the weight of 1 packet of sugar?
4 7/8 / 5 = 39/8 x 1/5 = 39/40 kg.

5. Ayesha's age is 1/2 of her sister's age. If her sister is 20 years old then how much old is Ayesha?
1/2 x 20 = 10 years old.

6. (a) Kamal will distribute 4 1/2 packets of candies among 6 children. How many packets of candies will each child get?
4 1/2 / 6 = 9/2 x 1/6 = 9/12 = 3/4 packet.

6. (b) If he distributes these packets among 9 children, then how many packets of candies will each child get?
4 1/2 / 9 = 9/2 x 1/9 = 9/18 = 1/2 packet.

7. The distance between Hamid's home and Masjid is 2 3/4 kilometre. If Hamid goes to Masjid to offer the Salah five times, then how much distance does he cover daily?
2 3/4 x 5 = 11/4 x 5 = 55/4 = 13 3/4 km.

## Review Exercise (book pages 68-69)

1. (a) _____ is a proper fraction.
(a) 5/4
(b) 9/7
(c) 1/2 ✓
(d) 4/2

1. (b) _____ is an improper fraction.
(a) 5/4 ✓
(b) 1/2
(c) 3 5/9
(d) 4/7

1. (c) 1/4 + 3/4 is equal to _____.
(a) 1/4
(b) 3/4
(c) 1 ✓

(d) 2/4

1. (d) 7/6 - 2/6 is equal to _____.
(a) 9/6
(b) 2/6
(c) 1/6
(d) 5/6 ✓

1. (e) The product of 7/6 and 5 is _____.
(a) 34/6
(b) 34/7
(c) 35/7
(d) 35/6 ✓

2. Tick the like fractions: (a) 4/9, 1/9, 2/9; (b) 5/7, 6/11, 2/13; (c) 2/17, 9/23, 11/12; (d) 6/14, 1/14, 2/14; (e) 4/13, 1/5, 6/7; (f) 3/10, 9/10, 1/10.
Like fractions (same denominator): (a) 4/9, 1/9, 2/9; (d) 6/14, 1/14, 2/14; (f) 3/10, 9/10, 1/10.

3. Compare and write <, > or =: (a) 9/10 __ 5/7; (b) 2/24 __ 1/12; (c) 6/11 __ 3/5; (d) 8/9 __ 1/2.
(a) 9/10 = 63/70 > 50/70 = 5/7. (b) 2/24 = 1/12. (c) 6/11 = 30/55 < 33/55 = 3/5. (d) 8/9 = 16/18 > 9/18 = 1/2.

4. Write into the lowest form: (a) 52/18; (b) 17/51; (c) 9/19; (d) 22/33.
(a) 52/18 = 26/9. (b) 17/51 = 1/3. (c) 9/19 is already in the lowest form. (d) 22/33 = 2/3.

5. Encircle the unit fractions and tick the improper fractions: (a) 9/9; (b) 1/6; (c) 7/2; (d) 11/5; (e) 1/9.
Unit fractions (numerator 1): 1/6, 1/9. Improper fractions (numerator equal to or greater than denominator): 9/9, 7/2, 11/5.

6. Convert improper fraction 7/5 into a mixed number.
7/5 = 1 2/5.

6. Convert improper fraction 11/8 into a mixed number.
11/8 = 1 3/8.

6. Convert improper fraction 17/4 into a mixed number.
17/4 = 4 1/4.

6. Convert improper fraction 5/4 into a mixed number.
5/4 = 1 1/4.

7. Convert mixed number 1 3/5 into an improper fraction.
1 3/5 = 8/5.

7. Convert mixed number 3 5/10 into an improper fraction.
3 5/10 = 35/10.

7. Convert mixed number 6 4/7 into an improper fraction.
6 4/7 = 46/7.

7. Convert mixed number 2 1/9 into an improper fraction.
2 1/9 = 19/9.

8. (a) Write 4/8, 5/2, 6/7, 1/6 in ascending and descending order.
Ascending: 1/6, 4/8, 6/7, 5/2. Descending: 5/2, 6/7, 4/8, 1/6.

8. (b) Write 2/9, 8/9, 5/6, 1/3 in ascending and descending order.
With denominator 18: 2/9 = 4/18, 8/9 = 16/18, 5/6 = 15/18, 1/3 = 6/18.
Ascending: 2/9, 1/3, 5/6, 8/9. Descending: 8/9, 5/6, 1/3, 2/9.

8. (c) Write 4/12, 7/18, 3/10, 5/6 in ascending and descending order.
With denominator 180: 4/12 = 60/180, 7/18 = 70/180, 3/10 = 54/180, 5/6 = 150/180. Ascending: 3/10, 4/12, 7/18, 5/6. Descending: 5/6, 7/18, 4/12, 3/10.

9. Add: (a) 1/3, 2/3; (b) 11/12, 7/12; (c) 9/5, 4/5; (d) 3/9, 7/9.
(a) 1/3 + 2/3 = 1. (b) 11/12 + 7/12 = 18/12 = 3/2 = 1 1/2. (c) 9/5 + 4/5 = 13/5 = 2 3/5. (d) 3/9 + 7/9 = 10/9 = 1 1/9.

10. Solve: (a) 9/10 - 3/10; (b) 8/13 - 2/13; (c) 6/17 - 1/17; (d) 4/7 - 3/7.
(a) 9/10 - 3/10 = 6/10 = 3/5. (b) 8/13 - 2/13 = 6/13. (c) 6/17 - 1/17 = 5/17. (d) 4/7 - 3/7 = 1/7.

11. (a) 9/5 x 7.
9/5 x 7 = 63/5 = 12 3/5.

11. (b) 4 2/11 x 9/10.
4 2/11 x 9/10 = 46/11 x 9/10 = 414/110 = 207/55 = 3 42/55.

11. (c) 1/2 x 7 8/9.
1/2 x 7 8/9 = 1/2 x 71/9 = 71/18 = 3 17/18.

11. (d) 3 2/11 x 1 5/14.
3 2/11 x 1 5/14 = 35/11 x 19/14 = 665/154 = 4 7/22.

11. (e) 1 1/4 x 7 3/9.
1 1/4 x 7 3/9 = 5/4 x 66/9 = 330/36 = 55/6 = 9 1/6.

11. (f) 1/6 / 9.
1/6 / 9 = 1/6 x 1/9 = 1/54.

11. (g) 4 7/12 / 1.
4 7/12 / 1 = 55/12 = 4 7/12.

12. In a garden 7/12 of the trees are mango trees. In another garden 9/12 of the trees are mango trees. How many mango trees are there altogether?
7/12 + 9/12 = 16/12 = 4/3 = 1 1/3 (in fraction of the trees).

13. Hania has 12 4/7 m of ribbon. She wants to cut it into 8 equal pieces. What will be the length of each piece?
12 4/7 / 8 = 88/7 x 1/8 = 88/56 = 11/7 = 1 4/7 m.

14. Jamal reads 2/7 of 140 pages of a book and Farhan reads 2 times more pages than Jamal. How many pages does Farhan read?
Jamal reads 2/7 x 140 = 40 pages. Farhan reads 2 x 40 = 80 pages.

## Unit 4 - Decimals

### Exercise 1 (book pages 75-76)

1. Write the given fractions in decimals: (a) 16/100, (b) 1/10, (c) 324/1000, (d) 2/100, (e) 70/1000.
Count the zeros in the denominator and place the decimal point that many digits from the right: (a) 16/100 = 0.16, (b) 1/10 = 0.1, (c) 324/1000 = 0.324, (d) 2/100 = 0.02, (e) 70/1000 = 0.070.

2. Write the place value of the coloured digits: (a) 1.56 (digit 5), (b) 45.987 (digit 8), (c) 321.17 (digit 3), (d) 6.897 (digit 9), (e) 6.340 (digit 0), (f) 12.123 (digit 2), (g) 78.808 (digit 8), (h) 4.009 (digit 0).
(a) 5 is at tenths place: 5 x 0.1 = 0.5. (b) 8 is at hundredths place: 8 x 0.01 = 0.08. (c) 3 is at hundreds place: 3 x 100 = 300. (d) 9 is at hundredths place: 9 x 0.01 = 0.09. (e) 0 is at thousandths place: 0 x 0.001 = 0.000. (f) 2 is at ones place: 2 x 1 = 2. (g) 8 is at thousandths place: 8 x 0.001 = 0.008. (h) 0 is at tenths place: 0 x 0.1 = 0.0.

3. Colour the shapes with the help of the given decimals: (a) 0.2, (b) 0.7, (c) 0.9. (Each shape is divided into 10 equal parts.)

Colour (a) 2 parts out of 10 for 0.2, (b) 7 parts out of 10 for 0.7, (c) 9 parts out of 10 for 0.9.

4. Fill in the blanks (place values): (a) 7.45, (b) 26.9, (c) 87.391, (d) 99.999.
(a) 7.45: 7 is at ones place, 7 x 1 = 7; 4 is at tenths place, 4 x 0.1 = 0.4; 5 is at hundredths place, 5 x 0.01 = 0.05. (b) 26.9: 2 is at tens place, 2 x 10 = 20; 6 is at ones place, 6 x 1 = 6; 9 is at tenths place, 9 x 0.1 = 0.9. (c) 87.391: 8 is at tens place, 8 x 10 = 80; 7 is at ones place, 7 x 1 = 7; 3 is at tenths place, 3 x 0.1 = 0.3; 9 is at hundredths place, 9 x 0.01 = 0.09; 1 is at thousandths place, 1 x 0.001 = 0.001. (d) 99.999: 9 is at tens place, 9 x 10 = 90; 9 is at ones place, 9 x 1 = 9; 9 is at tenths place, 9 x 0.1 = 0.9; 9 is at hundredths place, 9 x 0.01 = 0.09; 9 is at thousandths place, 9 x 0.001 = 0.009.

Try It! (book page 75): Write the place value of each digit in 200.149 and express in expanded form.
2 is at hundreds place: 2 x 100 = 200. 0 is at tens place: 0 x 10 = 0. 0 is at ones place: 0 x 1 = 0. 1 is at tenths place: 1 x 0.1 = 0.1. 4 is at hundredths place: 4 x 0.01 = 0.04. 9 is at thousandths place: 9 x 0.001 = 0.009. Expanded form: 200.149 = 200 + 0 + 0 + 0.1 + 0.04 + 0.009.

## Exercise 2 (book page 80)

1. Represent these into decimal: (a) 24/100, (b) 5/1000, (c) 6/10, (d) 12/500, (e) 3/250, (f) 47/25, (g) 60/200, (h) 606/1000, (i) 80/1000, (j) 1/50.
Convert each to denominator 10, 100 or 1000, then write the decimal: (a) 24/100 = 0.24, (b) 5/1000 = 0.005, (c) 6/10 = 0.6, (d) 12/500 = 24/1000 = 0.024, (e) 3/250 = 12/1000 = 0.012, (f) 47/25 = 188/100 = 1.88, (g) 60/200 = 30/100 = 0.3, (h) 606/1000 = 0.606, (i) 80/1000 = 0.08, (j) 1/50 = 2/100 = 0.02.

2. Convert the following decimals into fractions and write them in lowest form: (a) 1.3, (b) 2.04, (c) 6.98, (d) 5.5, (e) 0.68, (f) 21.72, (g) 7.87, (h) 0.98, (i) 11.11, (j) 6.10.
(a) 1.3 = 13/10. (b) 2.04 = 204/100 = 51/25. (c) 6.98 = 698/100 = 349/50. (d) 5.5 = 55/10 = 11/2. (e) 0.68 = 68/100 = 17/25. (f) 21.72 = 2172/100 = 543/25. (g) 7.87 = 787/100. (h) 0.98 = 98/100 = 49/50. (i) 11.11 = 1111/100. (j) 6.10 = 610/100 = 61/10.

Try It! (book page 79): Show 7.74 in the place value chart and convert 7.74 into a fraction.
7 is at ones place (7 x 1 = 7), 7 is at tenths place (7 x 0.1 = 0.7), 4 is at hundredths place (4 x 0.01 = 0.04). As a fraction: 7.74 = 774/100 = 387/50.

## Exercise 3 (book page 82)

1. Add: (a) 9.11, 8.03; (b) 43.1, 12.7; (c) 52.9, 2.2; (d) 5.69, 2.98; (e) 6.02, 1.89; (f) 49.3, 21.6.

(a) 9.11 + 8.03 = 17.14. (b) 43.1 + 12.7 = 55.8. (c) 52.9 + 2.2 = 55.1. (d) 5.69 + 2.98 = 8.67. (e) 6.02 + 1.89 = 7.91. (f) 49.3 + 21.6 = 70.9.

2. Solve: (a) 4.91 - 3.92; (b) 7.34 - 2.86; (c) 5.06 - 2.76; (d) 7.88 - 6.19; (e) 9.80 - 1.09; (f) 78.9 - 7.84.

(a) 4.91 - 3.92 = 0.99. (b) 7.34 - 2.86 = 4.48. (c) 5.06 - 2.76 = 2.30. (d) 7.88 - 6.19 = 1.69. (e) 9.80 - 1.09 = 8.71. (f) 78.9 - 7.84 = 71.06.

3. Zubair bought a chocolate for Rs 45.7 and a candy for Rs 10.2. How much amount did he spend altogether?

Amount spent = 45.7 + 10.2. Step 1: add tenths: 7 + 2 = 9 tenths. Step 2: add ones: 5 + 0 = 5, plus the carried 1 from 45 + 10 = 55. Total = Rs 55.9.

4. The mass of apples is 38.9 kg and mass of guava is 42.6 kg. (a) Find the difference between mass of apples and guavas. (b) Find the total mass.

(a) Difference = 42.6 - 38.9 = 3.7 kg. (b) Total mass = 38.9 + 42.6 = 81.5 kg.

Try It! (book page 82): If the sum of two decimals is 12.8 and the second decimal fraction is 7.9, find the first decimal fraction. What do we add in 12.8 to get 25.8?

First decimal = 12.8 - 7.9 = 4.9. To get 25.8 from 12.8 we add 25.8 - 12.8 = 13.0.

## Exercise 4 (book page 86)

1. Solve: (a) 5.9 x 10, (b) 4.8 x 100, (c) 0.3 x 1000, (d) 8.2 x 10, (e) 4.3 x 1000, (f) 9.1 x 100.

Move the decimal point right (1 place for 10, 2 for 100, 3 for 1000): (a) 5.9 x 10 = 59. (b) 4.8 x 100 = 480. (c) 0.3 x 1000 = 300. (d) 8.2 x 10 = 82. (e) 4.3 x 1000 = 4300. (f) 9.1 x 100 = 910.

2. Solve: (a) 5.6 x 8, (b) 7.1 x 2, (c) 4.9 x 4, (d) 3.4 x 3, (e) 1.3 x 7, (f) 9.8 x 9.

(a) 5.6 x 8 = 44.8. (b) 7.1 x 2 = 14.2. (c) 4.9 x 4 = 19.6. (d) 3.4 x 3 = 10.2. (e) 1.3 x 7 = 9.1. (f) 9.8 x 9 = 88.2.

3. Solve: (a) 1.4 / 2, (b) 1.8 / 9, (c) 6.4 / 4, (d) 8.4 / 6, (e) 2.7 / 3, (f) 2.6 / 2.

(a) 1.4 / 2 = 0.7. (b) 1.8 / 9 = 0.2. (c) 6.4 / 4 = 1.6. (d) 8.4 / 6 = 1.4. (e) 2.7 / 3 = 0.9. (f) 2.6 / 2 = 1.3.

4. Saba uses 9.8 ml of oil to bake a cake. How much oil will she use to bake

10 such cakes?
Oil for 10 cakes = 9.8 x 10 = 98 ml (decimal point moves 1 place right).

5. A tailor uses 2.5 m cloth to make a shirt. How much cloth will he use to make 8 similar shirts?
Cloth for 8 shirts = 2.5 x 8 = 20 m.

6. The length of one piece of rope is 7.2 m. Iram cuts this rope into 4 equal pieces. (a) What will be the length of each piece? (b) If she cuts the rope into 2 equal pieces, what will be the length of each piece?
(a) 7.2 / 4 = 1.8 m per piece. (b) 7.2 / 2 = 3.6 m per piece.

7. Ahmed solves 5 questions of Mathematics in 8.5 minutes. How long does he take to solve 1 question?
Time for 1 question = 8.5 / 5 = 1.7 minutes.

Try It! (book page 85): Divide 2.8 by 4. Write decimals with one decimal place which when divided by 4 will give 1.2, 0.2 and 0.4.
2.8 / 4 = 0.7. The decimals are: 1.2 x 4 = 4.8, 0.2 x 4 = 0.8, 0.4 x 4 = 1.6.

## Exercise 5 (book page 89)

1. Round off the following whole numbers to the nearest 10, 100 and 1000: (a) 9871, (b) 5467, (c) 1212, (d) 6343, (e) 5555, (f) 3498, (g) 1289, (h) 4545, (i) 1111.
(a) 9871: 9870, 9900, 10000. (b) 5467: 5470, 5500, 5000. (c) 1212: 1210, 1200, 1000. (d) 6343: 6340, 6300, 6000. (e) 5555: 5560, 5600, 6000. (f) 3498: 3500, 3500, 3000. (g) 1289: 1290, 1300, 1000. (h) 4545: 4550, 4500, 5000. (i) 1111: 1110, 1100, 1000.

2. Round off the following decimal fractions to the nearest whole number: (a) 5.61, (b) 54.2, (c) 987.4, (d) 12.7, (e) 8.98, (f) 6.5, (g) 76.4, (h) 8.19, (i) 87.87.
If the digit after the decimal point is 5 or more, round up; otherwise round down: (a) 5.61 = 6. (b) 54.2 = 54. (c) 987.4 = 987. (d) 12.7 = 13. (e) 8.98 = 9. (f) 6.5 = 7. (g) 76.4 = 76. (h) 8.19 = 8. (i) 87.87 = 88.

Try It! (book page 89): Write 3 decimals which can be rounded off to 68.
Model answers: 67.5, 67.9 and 68.2 (each rounds to 68 to the nearest whole number).

## Review Exercise (book pages 91-92)

1. (a) Decimal is a fraction with the denominator is power of _____.
(a) 10
(b) 2

(c) 15
(d) 0

1. (b) When we divide a shape into 10 equal parts, then each part is called _____.
(a) hundredths
(b) tenths
(c) one
(d) half

1. (c) To add the decimals always _____ ones in ones, tenths in tenths and hundredths in hundredths.
(a) add
(b) subtract
(c) multiply
(d) divide

1. (d) When we multiply any decimal by 100, we move the decimal _____ place to the right.
(a) 1
(b) 2
(c) 3
(d) 0

1. (e) _____ means to find a number that is nearest to the original number but not exact.
(a) Decimal
(b) Fraction
(c) Round off
(d) Estimation

2. Represent the following in decimals: (a) 17/100, (b) 15/200, (c) 4/20, (d) 19/100, (e) 200/250.
(a) 17/100 = 0.17. (b) 15/200 = 75/1000 = 0.075. (c) 4/20 = 20/100 = 0.20. (d) 19/100 = 0.19. (e) 200/250 = 800/1000 = 0.800.

3. Convert the following decimals into fractions: (a) 6.7, (b) 45.56, (c) 1.02, (d) 7.87, (e) 15.8.
(a) 6.7 = 67/10. (b) 45.56 = 4556/100 = 1139/25. (c) 1.02 = 102/100 = 51/50. (d) 7.87 = 787/100. (e) 15.8 = 158/10 = 79/5.

4. Add the following: (a) 6.03, 5.56; (b) 8.28, 1.24; (c) 12.8, 3.14.
(a) 6.03 + 5.56 = 11.59. (b) 8.28 + 1.24 = 9.52. (c) 12.8 + 3.14 = 15.94.

5. Solve: (a) 7.59 - 2.48, (b) 6.19 - 4.21, (c) 5.06 - 1.09.
(a) 7.59 - 2.48 = 5.11. (b) 6.19 - 4.21 = 1.98. (c) 5.06 - 1.09 = 3.97.

6. Solve: (a) 1.3 x 10, (b) 8.9 x 100, (c) 4.5 x 1000, (d) 8.2 x 7, (e) 9.3 x 3, (f) 7.1 x 6.
(a) 1.3 x 10 = 13. (b) 8.9 x 100 = 890. (c) 4.5 x 1000 = 4500. (d) 8.2 x 7 = 57.4. (e) 9.3 x 3 = 27.9. (f) 7.1 x 6 = 42.6.

7. Solve: (a) 6.8 / 4, (b) 8.8 / 2, (c) 8.1 / 9, (d) 3.5 / 7, (e) 4.8 / 6, (f) 5.1 / 3.
(a) 6.8 / 4 = 1.7. (b) 8.8 / 2 = 4.4. (c) 8.1 / 9 = 0.9. (d) 3.5 / 7 = 0.5. (e) 4.8 / 6 = 0.8. (f) 5.1 / 3 = 1.7.

8. Round off the whole number to the nearest 10, 100 and 1000: (a) 3429, (b) 1009, (c) 7824, (d) 8417, (e) 4090, (f) 1717.
(a) 3429: 3430, 3400, 3000. (b) 1009: 1010, 1000, 1000. (c) 7824: 7820, 7800, 8000. (d) 8417: 8420, 8400, 8000. (e) 4090: 4090, 4100, 4000. (f) 1717: 1720, 1700, 2000.

9. Round off the decimals to the nearest whole number: (a) 4.17, (b) 78.3, (c) 13.45, (d) 0.98, (e) 41.41, (f) 9.82.
(a) 4.17 = 4. (b) 78.3 = 78. (c) 13.45 = 13. (d) 0.98 = 1. (e) 41.41 = 41. (f) 9.82 = 10.

10. The length of a wire is 3.41 m and length of another wire is 7.56 m. (a) What will be the total length? (b) What is the difference between the lengths of the wires?
(a) Total length = 3.41 + 7.56 = 10.97 m. (b) Difference = 7.56 - 3.41 = 4.15 m.

11. The capacity of a pack of juice is 3.4 litre. (a) What will be the capacity of such 7 packs? (b) If the capacity of a pack of juice is 2.8 litre, what will be the capacity of 5 such packs?
(a) 3.4 x 7 = 23.8 litres. (b) 2.8 x 5 = 14 litres.

12. The mass of 4 boxes of pencils is 1.8 kg. What will be the mass of 1 box?
Mass of 1 box = 1.8 / 4 = 0.45 kg.

# Unit 5 - Measurement

## Exercise 1 (book page 97)

1. Tick the correct units of length for the pictured objects: (a), (b), (c), (d).
The choices are m/cm, mm/cm, m/cm, m/km.

The correct units for the four pictured objects are: (a) cm, (b) cm, (c) m, (d) km (a short object is measured in cm, a long object in m, and a distance between places in km).

2. Convert these units of length: (a) 12 km into m, (b) 56 km 930 m into m, (c) 88 m into cm, (d) 60 m 78 cm into cm, (e) 3.2 cm into mm, (f) 55 cm 2 mm into mm.

(a) 12 km = 12 x 1000 = 12000 m. (b) 56 km 930 m = 56000 + 930 = 56930 m. (c) 88 m = 88 x 100 = 8800 cm. (d) 60 m 78 cm = 6000 + 78 = 6078 cm. (e) 3.2 cm = 3.2 x 10 = 32 mm. (f) 55 cm 2 mm = 550 + 2 = 552 mm.

## Exercise 2 (book page 99)

1. Solve: (a) 22 km + 33 km, (b) 88 km + 6 km 17 m, (c) 82 m + 22 m, (d) 71 cm 2 mm + 11 cm 6 mm, (e) 21 m 16 cm + 20 m 14 cm, (f) 74 km 122 m + 13 m.

(a) 22 + 33 = 55 km. (b) 88 km + 6 km 17 m = 94 km 17 m. (c) 82 + 22 = 104 m. (d) 71 cm 2 mm + 11 cm 6 mm = 82 cm 8 mm. (e) 21 m 16 cm + 20 m 14 cm = 41 m 30 cm. (f) 74 km 122 m + 13 m = 74 km 135 m.

2. Solve: (a) 99 km - 74 km, (b) 52 km 48 m - 6 km 22 m, (c) 47 m - 10 m, (d) 35 cm 5 mm - 25 cm 1 mm, (e) 21 m 16 cm - 20 m 14 cm, (f) 74 km 122 m - 13 m.

(a) 99 - 74 = 25 km. (b) 52 km 48 m - 6 km 22 m = 46 km 26 m. (c) 47 - 10 = 37 m. (d) 35 cm 5 mm - 25 cm 1 mm = 10 cm 4 mm. (e) 21 m 16 cm - 20 m 14 cm = 1 m 2 cm. (f) 74 km 122 m - 13 m = 74 km 109 m.

3. Tahir has two pieces of rope. The length of one piece is 38 m 87 cm and length of the other piece is 61 m 12 cm. What will be the total length?

Total length = 38 m 87 cm + 61 m 12 cm. Step 1: add cm: 87 + 12 = 99 cm. Step 2: add m: 38 + 61 = 99 m. Total = 99 m 99 cm.

4. Ahmed buys 140 cm ribbon to wrap the gift box. Convert the length into millimetres.

140 cm = 140 x 10 = 1400 mm.

5. Kamal covers distance of 2 km 712 m to go from school to home. He covers distance of 1 km 216 m to go from home to Masjid. (a) Find the difference between the distances. (b) Convert the difference into metres.

(a) Difference = 2 km 712 m - 1 km 216 m = 1 km 496 m. (b) 1 km 496 m = 1000 + 496 = 1496 m.

6. The length of Ahmed's room is 5 m 56 cm and his sister's room is 4 m 44 cm. (a) What will be the total length of both rooms in cm? (b) What is the

difference between the length of both rooms?

(a) Total = 5 m 56 cm + 4 m 44 cm = 9 m 100 cm = 10 m = 1000 cm. (b) Difference = 5 m 56 cm - 4 m 44 cm = 1 m 12 cm.

## Exercise 3 (book pages 104-105)

1. Convert the following units of mass: (a) 65 kg into g, (b) 23 kg 139 g into g, (c) 89 g into mg, (d) 43 g 699 mg into mg, (e) 1.9 g into mg, (f) 0.8 kg into g.

(a) 65 kg = 65 x 1000 = 65000 g. (b) 23 kg 139 g = 23000 + 139 = 23139 g. (c) 89 g = 89 x 1000 = 89000 mg. (d) 43 g 699 mg = 43000 + 699 = 43699 mg. (e) 1.9 g = 1.9 x 1000 = 1900 mg. (f) 0.8 kg = 0.8 x 1000 = 800 g.

2. Solve: (a) 36 kg + 76 kg, (b) 18 kg + 17 kg 17 g, (c) 8.2 g + 2.2 g, (d) 71 g 2 mg + 11 g 560 mg, (e) 21 g 16 mg + 60 g 14 mg, (f) 94 kg 122 g + 23 g.

(a) 36 + 76 = 112 kg. (b) 18 kg + 17 kg 17 g = 35 kg 17 g. (c) 8.2 + 2.2 = 10.4 g. (d) 71 g 2 mg + 11 g 560 mg = 82 g 562 mg. (e) 21 g 16 mg + 60 g 14 mg = 81 g 30 mg. (f) 94 kg 122 g + 23 g = 94 kg 145 g.

3. Solve: (a) 99 kg - 24 kg, (b) 58 kg 458 g - 29 kg 303 g, (c) 904 g - 154 g, (d) 39 g 500 mg - 25 g 100 mg.

(a) 99 - 24 = 75 kg. (b) 58 kg 458 g - 29 kg 303 g = 29 kg 155 g. (c) 904 - 154 = 750 g. (d) 39 g 500 mg - 25 g 100 mg = 14 g 400 mg.

4. Usman has two fish in a jar. The mass of one fish is 29 g 20 mg and the mass of the other is 20 g 14 mg. What is the difference between the masses of the two fish in mg?

Difference = 29 g 20 mg - 20 g 14 mg = 9 g 6 mg. In mg: 9 x 1000 + 6 = 9006 mg.

5. A shopkeeper sells 49 kg 208 g of sugar and 65 kg 750 g of flour. Find the total quantity of sugar and flour in grams.

Total = 49 kg 208 g + 65 kg 750 g. Step 1: add g: 208 + 750 = 958 g. Step 2: add kg: 49 + 65 = 114 kg. Total = 114 kg 958 g = 114958 g. (The difference between the two masses is 65 kg 750 g - 49 kg 208 g = 16 kg 542 g = 16542 g.)

## Exercise 4 (book page 107)

1. Convert the following units of capacities: (a) 9 l into ml, (b) 74 l into ml, (c) 56 l into ml, (d) 90 l into ml, (e) 67 l into ml, (f) 1.6 l into ml.

(a) 9 l = 9000 ml. (b) 74 l = 74000 ml. (c) 56 l = 56000 ml. (d) 90 l = 90000 ml. (e) 67 l = 67000 ml. (f) 1.6 l = 1600 ml.

2. The capacity of an oil tanker is 98 l. Convert it into millilitres.
98 l = 98 x 1000 = 98000 ml.

3. Faria uses 1.7 litres of milk to make milkshake. Convert the quantity of milk into millilitres.
1.7 l = 1.7 x 1000 = 1700 ml.

## Exercise 5 (book page 109)

1. Solve: (a) 3 l 109 ml + 5 l 304 ml, (b) 6.5 l + 4.2 l, (c) 122 ml + 76 ml, (d) 34 l 200 ml + 92 l, (e) 41 l 200 ml + 404 l 478 ml.
(a) 3 l 109 ml + 5 l 304 ml = 8 l 413 ml. (b) 6.5 + 4.2 = 10.7 l. (c) 122 + 76 = 198 ml. (d) 34 l 200 ml + 92 l = 126 l 200 ml. (e) 41 l 200 ml + 404 l 478 ml = 445 l 678 ml.

2. Solve: (a) 22 l 500 ml - 10 l 109 ml, (b) 55 l - 32 l, (c) 2.2 ml - 1.5 ml, (d) 4 l 878 ml - 3 l 760 ml, (e) 78 l 209 ml - 16 l 142 ml.
(a) 22 l 500 ml - 10 l 109 ml = 12 l 391 ml. (b) 55 - 32 = 23 l. (c) 2.2 - 1.5 = 0.7 ml. (d) 4 l 878 ml - 3 l 760 ml = 1 l 118 ml. (e) 78 l 209 ml - 16 l 142 ml = 62 l 67 ml.

3. Zara has two containers. The capacity of one container is 67 l 198 ml and the other is 84 l 300 ml. (a) What is the total capacity of the containers? (b) What is the difference between the capacity of both containers?
(a) Total = 67 l 198 ml + 84 l 300 ml = 151 l 498 ml. (b) Difference = 84 l 300 ml - 67 l 198 ml = 17 l 102 ml.

4. A shopkeeper sells 72.8 l milk on Saturday and 92.6 l milk on Sunday. On which day does he sell less milk and how much? Give your answer in millilitres.
He sells less milk on Saturday. Difference = 92.6 - 72.8 = 19.8 l = 19800 ml.

## Review Exercise - Measurement (book pages 110-111)

1. (a) There are _____ metres in one kilometre.
(a) 1
(b) 10
(c) 100
(d) 1000 ✓

1. (b) There are _____ grams in one kilogram.
(a) 1
(b) 10
(c) 100
(d) 1000 ✓

1. (c) To convert cm into mm, multiply it by _____.
(a) 10000
(b) 100
(c) 10
(d) 1000

1. (d) One metre is equal to _____ centimetres.
(a) 1
(b) 10
(c) 1000
(d) 100

1. (e) One litre is equal to 1000 _____.
(a) grams
(b) metres
(c) millilitres
(d) litres

2. Convert the given units: (a) 105 km into m, (b) 3.4 m into cm, (c) 66 kg into g, (d) 60 g 498 mg into mg, (e) 0.5 g into mg, (f) 76 l 2 ml into ml, (g) 9.8 l into ml, (h) 90 cm into mm.
(a) 105 km = 105000 m. (b) 3.4 m = 340 cm. (c) 66 kg = 66000 g. (d) 60 g 498 mg = 60498 mg. (e) 0.5 g = 500 mg. (f) 76 l 2 ml = 76002 ml. (g) 9.8 l = 9800 ml. (h) 90 cm = 900 mm.

3. Solve: (a) 5 kg 299 g + 9 kg, (b) 4.5 m + 2.8 m, (c) 78 kg 700 g + 81 kg 172 g, (d) 212 l 200 ml + 92 l 500 ml, (e) 119 l 329 ml + 365 l 262 ml, (f) 65 g 313 mg + 98 g 198 mg.
(a) 5 kg 299 g + 9 kg = 14 kg 299 g. (b) 4.5 + 2.8 = 7.3 m. (c) 78 kg 700 g + 81 kg 172 g = 159 kg 872 g. (d) 212 l 200 ml + 92 l 500 ml = 304 l 700 ml. (e) 119 l 329 ml + 365 l 262 ml = 484 l 591 ml. (f) 65 g 313 mg + 98 g 198 mg = 163 g 511 mg.

4. Solve: (a) 506 kg - 417 kg, (b) 85 m 405 cm - 13 m 21 cm, (c) 84 kg 711 g - 75 kg 680 g, (d) 78 l 805 ml - 63 l 214 ml, (e) 98 l 788 ml - 45 l 659 ml.
(a) 506 - 417 = 89 kg. (b) 85 m 405 cm - 13 m 21 cm = 72 m 384 cm. (c) 84 kg 711 g - 75 kg 680 g = 9 kg 31 g. (d) 78 l 805 ml - 63 l 214 ml = 15 l 591 ml. (e) 98 l 788 ml - 45 l 659 ml = 53 l 129 ml.

5. The height of K-2 is 8 km 611 m and Mount Everest is 8 km 848 m. (a) What is the difference between their heights? Give your answer in metres. (b) Find the total height of the mountains?
(a) Difference = 8 km 848 m - 8 km 611 m = 237 m. (b) Total = 8611 + 8848 = 17459 m = 17 km 459 m.

6. On Eid, Asim buys a goat and a camel to sacrifice. The goat weighs 55 kg and the camel weighs 200 kg. (a) Find the total mass of both animals. (b) Convert the total mass into grams.
(a) Total mass = 55 + 200 = 255 kg. (b) 255 kg = 255000 g.

7. The capacity of a bucket is 87 l and a tub is 112 l 456 ml. What is the total capacity in millilitres?
Total = 87 l + 112 l 456 ml = 199 l 456 ml = 199456 ml.

## Unit 5 - Time

### Exercise 1 - Time (book pages 116-117)

1. Colour the correct time of your daily activities using the 12 hours clock format: (a) Getting up for the school, (b) Going to the school, (c) Lunch time in the school, (d) Playing time in the evening, (e) Isha Salah, (f) Sleeping time at night.
The correct 12-hour times are: (a) 6:00 a.m. (getting up), (b) 7:45 a.m. (going to school), (c) 12:30 p.m. (lunch), (d) 4:30 p.m. (playing in the evening), (e) 8:00 p.m. (Isha Salah), (f) 10:25 p.m. (sleeping at night).

2. With the help of clocks, write the correct time in 24-hour format: (a) When does Numan take his breakfast? (b) When does Farheen play with her friends? (c) When does Junaid go to sleep?
Reading the three clock faces in 24-hour format: (a) 7:30:15, (b) 5:30:35, (c) 8:59:58.

3. Look at the following clocks and tell the time in hours, minutes and seconds: (a), (b), (c), (d).
Reading the four clock faces: (a) 5:00:15, (b) 5:00:35, (c) 7:40:00, (d) 3:10:00.

4. Draw hour, minute and second hands according to the given time: (a) 2:15:30, (b) 6:25:03, (c) 10:22:55, (d) 9:11:31, (e) 8:20:01, (f) 12:42:16.
Draw the hands on each clock face at the given time: hour hand between the hour numbers, minute hand on the minutes, second hand on the seconds — (a) 2:15:30, (b) 6:25:03, (c) 10:22:55, (d) 9:11:31, (e) 8:20:01, (f) 12:42:16.

### Exercise 2 - Time (book page 120)

1. Convert the following time into minutes: (a) 6h, (b) 201h, (c) 14h 12min, (d) 5h 55min, (e) 17h 28min, (f) 22h 26min, (g) 11h 48min, (h) 22h 15min,

(i) 9h 43min, (j) 18h 6min, (k) 27h 38min, (l) 42h 26min.
Multiply hours by 60 and add the minutes: (a) 360; (b) 12060; (c) 852; (d) 355; (e) 1048; (f) 1346; (g) 708; (h) 1335; (i) 583; (j) 1086; (k) 1658; (l) 2546.

2. Convert the following time into seconds: (a) 77min, (b) 43min, (c) 1min 13sec, (d) 8min 32sec, (e) 6min 53sec, (f) 63min 58sec, (g) 88min 59sec, (h) 65min 37sec, (i) 214min 24sec, (j) 100min 11sec, (k) 176min 18sec, (l) 432min 03sec.
Multiply minutes by 60 and add the seconds: (a) 4620; (b) 2580; (c) 73; (d) 512; (e) 413; (f) 3838; (g) 5339; (h) 3937; (i) 12864; (j) 6011; (k) 10578; (l) 25923.

## Exercise 3 - Time (book pages 122-123)

1. Convert the following into months: (a) 9 years, (b) 4 years, (c) 12 years, (d) 21 years, (e) 8 years 3 months, (f) 5 years 4 months, (g) 20 years 6 months, (h) 17 years 10 months, (i) 30 years 11 months, (j) 29 years 7 months, (k) 15 years 11 months, (l) 43 years 7 months.
Multiply years by 12 and add the months (1 year = 12 months): (a) 108; (b) 48; (c) 144; (d) 252; (e) 99; (f) 64; (g) 246; (h) 214; (i) 371; (j) 355; (k) 191; (l) 523.

2. Convert the following into days: (a) 11 weeks, (b) 8 weeks, (c) 5 weeks, (d) 25 weeks, (e) 9 weeks 6 days, (f) 7 weeks 1 day, (g) 2 weeks 5 days, (h) 4 weeks 3 days, (i) 32 weeks 4 days, (j) 27 months 3 days, (k) 41 months 12 days, (l) 54 months 13 days.
Use 1 week = 7 days and 1 month = 30 days: (a) 77; (b) 56; (c) 35; (d) 175; (e) 69; (f) 50; (g) 19; (h) 31; (i) 228; (j) 813; (k) 1242; (l) 1633.

## Exercise 4 - Time (book page 125)

1. Solve: (a) 34h 11min 13sec + 11h 18min 32sec, (b) 24h 34min 37sec + 2h 21min 11sec, (c) 54h 19min 45sec + 43h 20min 10sec, (d) 5h 15min 31sec + 4h 4min 25sec, (e) 14 years 7 months 2 days + 7 years 4 months 2 days, (f) 51h 02min 08sec + 37h 11min 09sec, (g) 49 years 2 months 5 days + 40 years 5 months 11 days, (h) 27 years 3 months 5 days + 32 years 6 months 4 days.
(a) 45h 29min 45sec. (b) 26h 55min 48sec. (c) 97h 39min 55sec. (d) 9h 19min 56sec. (e) 14 + 7 = 21 years, 7 + 4 = 11 months, 2 + 2 = 4 days: 21 years 11 months 4 days. (f) 88h 13min 17sec. (g) 49 + 40 = 89 years, 2 + 5 = 7 months, 5 + 11 = 16 days: 89 years 7 months 16 days. (h) 27 + 32 = 59 years, 3 + 6 = 9 months, 5 + 4 = 9 days: 59 years 9 months 9 days.

2. Solve: (a) 45h 45min 49sec - 10h 23min 38sec, (b) 57h 22min 27sec - 33h 11min 12sec, (c) 65h 28min 56sec - 54h 20min 45sec, (d) 6h 26min

42sec - 5h 15min 31sec, (e) 25 years 8 months 3 days - 23 years 6 months 1 day, (f) 62h 53min 29sec - 51h 42min 08sec, (g) 88 years 11 months 29 days - 46 years 10 months 15 days, (h) 37 years 6 months 29 days - 17 years 6 months 18 days.

(a) 35h 22min 11sec. (b) 24h 11min 15sec. (c) 11h 8min 11sec. (d) 1h 11min 11sec. (e) 25 - 23 = 2 years, 8 - 6 = 2 months, 3 - 1 = 2 days: 2 years 2 months 2 days. (f) 11h 11min 21sec. (g) 88 - 46 = 42 years, 11 - 10 = 1 month, 29 - 15 = 14 days: 42 years 1 month 14 days. (h) 37 - 17 = 20 years, 6 - 6 = 0 months, 29 - 18 = 11 days: 20 years 0 months 11 days.

3. Ahmed went to his grandmother's home on Sunday and he stays there for 2 hours and 20 minutes. On Monday, he went to his aunt's home and he spent 4 hours and 23 minutes. Find: (a) How much time did he spend at his relatives' home? (b) Write the time in minutes.

(a) Total time = 2h 20min + 4h 23min = 6h 43min. (b) 6h 43min = 6 x 60 + 43 = 403 minutes.

4. Ejaz travelled 6 hours 34 minutes 45 seconds in a bus and 4 hours 20 minutes 12 seconds in a train. Find: (a) How much more time did he travel in the bus than the train? (b) How much did he travel altogether?

(a) More time in bus = 6h 34min 45sec - 4h 20min 12sec = 2h 14min 33sec.

(b) Total travel = 6h 34min 45sec + 4h 20min 12sec = 10h 54min 57sec.

5. Maha takes 9 hours 23 minutes to complete a picture while Rohan takes 7 hours 10 minutes to complete the same picture. Find: (a) How much more time does Maha take? (b) The total time they take altogether.

(a) Maha takes 9h 23min - 7h 10min = 2h 13min more. (b) Total = 9h 23min + 7h 10min = 16h 33min.

## Review Exercise - Time (book pages 126-128)

1. (a) There is 13:50 in 24 hour clock, what time will be in 12-hour clock?

(a) 1:50 a.m.

(b) 3:50 p.m.

(c) 1:50 p.m.

(d) 12:50 p.m. ✓

1. (b) There is 3:55 p.m. in 12 hour clock, what time will be in 24-hour clock?

(a) 13:55

(b) 14:55

(c) 15:55 ✓

(d) 16:55

1. (c) Which time is the longest from the following?

(a) 2 years
(b) 12 months
(c) 1 year 3 months
(d) 350 days

1. (d) There are _____ months in 2 years 6 months.
(a) 21
(b) 28
(c) 26
(d) 30

1. (e) To convert years into months, we multiply the given years by _____.
(a) 24
(b) 10
(c) 11
(d) 12

2. Colour the correct time boxes by using the 12-hour format: (a) Time of Fajar Salah, (b) Breakfast time, (c) School's assembly time, (d) Break time in school, (e) Time to watch TV in the evening, (f) Dinner time.
The correct 12-hour times are: (a) 5:00 a.m., (b) 7:30 a.m., (c) 8:00 a.m., (d) 12:20 p.m., (e) 6:00 p.m., (f) 7:25 p.m.

3. Convert the following into minutes: (a) 8h, (b) 112h, (c) 15h 13min, (d) 7h 15min, (e) 28h 39min, (f) 33h 25min, (g) 11h 13min, (h) 23h 16min, (i) 6h 54min.
Multiply hours by 60 and add the minutes: (a) 480; (b) 6720; (c) 913; (d) 435; (e) 1719; (f) 2005; (g) 673; (h) 1396; (i) 414.

4. Convert the following into seconds: (a) 78min, (b) 33min, (c) 4min 17sec, (d) 9min 42sec, (e) 2min 53sec, (f) 21min 11sec, (g) 97min 47sec, (h) 48min 47sec, (i) 433min 44sec.
Multiply minutes by 60 and add the seconds: (a) 4680; (b) 1980; (c) 257; (d) 582; (e) 173; (f) 1271; (g) 5867; (h) 2927; (i) 26024.

5. Convert the following into months: (a) 5 years, (b) 3 years, (c) 22 years, (d) 32 years, (e) 9 years 4 months, (f) 15 years 10 months, (g) 29 years 8 months, (h) 19 years 11 months, (i) 54 years 9 months.
Multiply years by 12 and add the months: (a) 60; (b) 36; (c) 264; (d) 384; (e) 112; (f) 190; (g) 356; (h) 239; (i) 657.

6. Convert the following into days: (a) 15 weeks, (b) 9 weeks, (c) 6 weeks, (d) 27 weeks, (e) 11 weeks 2 days, (f) 9 weeks 5 days, (g) 11 weeks 6 days, (h) 56 weeks 4 days, (i) 43 weeks 3 days.

Use 1 week = 7 days: (a) 105; (b) 63; (c) 42; (d) 189; (e) 79; (f) 68; (g) 83; (h) 396; (i) 304.

7. Solve: (a) 37h 11min 38sec + 32h 34min 16sec, (b) 44h 25min 38sec + 32h 34min 06sec, (c) 6h 25min 35sec + 3h 13min 14sec.
(a) 37h 11min 38sec + 32h 34min 16sec = 69h 45min 54sec. (b) 44h 25min 38sec + 32h 34min 06sec = 76h 59min 44sec. (c) 6h 25min 35sec + 3h 13min 14sec = 9h 38min 49sec.

8. Solve: (a) 38h 33min 38sec - 00h 22min 26sec, (b) 56h 56min 58sec - 44h 43min 37sec, (c) 37h 54min 49sec - 21h 32min 00sec.
(a) 38h 33min 38sec - 0h 22min 26sec = 38h 11min 12sec. (b) 56h 56min 58sec - 44h 43min 37sec = 12h 13min 21sec. (c) 37h 54min 49sec - 21h 32min 00sec = 16h 22min 49sec.

9. Afnan spends 5 hours 23 minutes for studying and 1 hour 20 minutes for playing. (a) How much time does he spend in both activities? (b) Write the total time in minutes.
(a) Total = 5h 23min + 1h 20min = 6h 43min. (b) 6h 43min = 6 x 60 + 43 = 403 minutes.

10. A cook takes 2 hours 43 minutes 54 seconds to make a savoury dish and 1 hour 12 minutes 24 seconds to make a dessert. Find how much more time does he take to make the savoury dish.
More time = 2h 43min 54sec - 1h 12min 24sec = 1h 31min 30sec.

11. Asfand Yar spends 7 years 4 months 2 days in Saudi Arabia and 2 years 7 months 7 days in UAE. (a) How much time does he spend out of the country? (b) Write the total time in days.
(a) Total = 7 + 2 = 9 years, 4 + 7 = 11 months, 2 + 7 = 9 days: 9 years 11 months 9 days. (b) In days (1 year = 12 months, 1 month = 30 days): 9 x 360 + 11 x 30 + 9 = 3240 + 330 + 9 = 3579 days.

12. Farheen completes her medical education in 4 years 10 months 7 days and her house job in 2 years 2 days. How much time did she spend in medical education and house job?
Total = 4 + 2 = 6 years, 10 + 0 = 10 months, 7 + 2 = 9 days: 6 years 10 months 9 days.

# Unit 6 - Geometry

## Exercise 1 (book page 133)

1. Identify parallel and non-parallel lines from the given lines and write their names as well: (a), (b), (c), (d), (e), (f).

Parallel lines (never meet, distance always the same): (a), (c), (e).

Non-parallel lines (meet or would meet if extended): (b), (d), (f).

## Exercise 2 (book page 141)

1. Write the type of angles in the following: (a), (b), (c), (d).

(a) Acute angle (less than 90 degrees). (b) Right angle (equal to 90 degrees). (c) Obtuse angle (greater than 90 degrees but less than 180 degrees). (d) Acute angle (less than 90 degrees).

2. Measure the following angles with a protractor. [Figure activity: the angles are printed figures; measure each with a protractor and write its measure.]

Place the protractor's centre point on the vertex and its baseline on the initial ray, then read the angle on the correct scale. The measures are read from the printed figures with a protractor (no text values are printed in the book or its answer list).

3. Construct the angles of the given measurements: (a) 60 degrees, (b) 30 degrees, (c) 115 degrees.

Draw a ray, place the protractor's centre point on its end point and the baseline on the ray, mark the required degree on the scale, remove the protractor and join the mark to the end point: (a) 60 degrees, (b) 30 degrees, (c) 115 degrees.

4. Measure the angles in the following figures and write their names as well. [Figure activity: measure each printed angle with a protractor, then name it acute, right or obtuse by its measure.]

An angle less than 90 degrees is acute, equal to 90 degrees is right, and greater than 90 degrees (but less than 180 degrees) is obtuse. Measure each printed figure with a protractor and name it by this rule.

## Exercise 3 (book page 145)

1. Draw the parts of the following circles: diameter of a circle, radius of a circle, centre of a circle.

Centre: the point in the middle of the circle, equally distant from every point on the circle. Radius: a line segment joining the centre to any point on the circle. Diameter: a line segment joining two points on the circle and passing through the centre (diameter = 2 x radius).

2. Write the name of parts of the circles: (a), (b), (c).

(a) Centre = R, radius = PR, RQ, RS, diameter = PQ. (b) Centre = L, radius = LJ, LK, diameter = JK. (c) Centre = C, radius = CD, AC, CB, diameter = AB.

## Exercise 4 (book page 148)

1. Find the perimeter of the given figures (square grid): (a) 5 m x 5 m square, (b) 7 cm x 4 cm rectangle, (c) 6 m x 4 m rectangle.

Perimeter = total length of the boundary (count the grid sides): (a) 5 + 5 + 5 + 5 = 20 m. (b) 7 + 4 + 7 + 4 = 22 cm. (c) 6 + 4 + 6 + 4 = 20 m.

2. Find the area of the given figures (square grid): (a) 2 x 2 squares (cm), (b) 12 x 7 squares (m), (c) 4 x 2 squares (cm), (d) 1 square (m).

Area = number of squares covered: (a) 2 x 2 = 4 cm^2. (b) 12 x 7 = 84 m^2. (c) 4 x 2 = 8 cm^2. (d) 1 m^2.

## Exercise 5 (book page 152)

1. Mark the figures where you can see a line of symmetry: (a), (b), (c), (d).

The figures with a line of symmetry (the line divides the figure into two equal mirror halves) are (a) and (b).

2. Complete the given figures with respect to their line of symmetry.

Draw the mirror half on the other side of the line of symmetry so that both halves match exactly (same shape, same size, reversed).

## Exercise 6 (book pages 155-156)

1. Write the names of these figures and label their vertices, edges, and surfaces: (a), (b), (c), (d), (e), (f).

Use the properties taught in the lesson: cube — 8 vertices, 12 edges, 6 (square) surfaces; cuboid — 8 vertices, 12 edges, 6 (rectangular) surfaces; sphere — 0 vertices, 0 edges, 1 curved surface; cylinder — 0 vertices, 2 edges, 3 surfaces (1 curved, 2 plane circular); cone — 1 vertex, 1 edge, 2 surfaces (1 curved, 1 plane circular); pyramid (square base) — 5 vertices, 8 edges, 5 surfaces (1 square, 4 triangular). Name each printed figure and label its vertices, edges and surfaces with these counts.

2. Mark the figure which has the given properties: (a) 0 vertices, 3 surfaces, 2 edges; (b) 8 vertices, 6 flat surfaces, 12 edges; (c) 0 vertices, 1 surface, 0 edges; (d) 5 vertices, 5 surfaces, 8 edges; (e) 1 vertex, 2 surfaces, 1 edge; (f) 8 vertices, 6 rectangular surfaces, 12 edges.

(a) Cylinder (0 vertices, 3 surfaces, 2 edges). (b) Cube (8 vertices, 6 flat square surfaces, 12 edges). (c) Sphere (0 vertices, 1 curved surface, 0 edges). (d) Pyramid (5 vertices, 5 surfaces, 8 edges). (e) Cone (1 vertex, 2 surfaces, 1 edge). (f) Cuboid (8 vertices, 6 rectangular surfaces, 12 edges).

## Review Exercise (book pages 158-161)

1. (a) The lines which keep going straight and never meet each other are called:

(a) horizontal lines
(b) vertical lines
(c) non-parallel lines
(d) parallel lines ✓

1. (b) There are _____ small parts in a protractor and each part is equal to 1 degree.
(a) 150
(b) 180 ✓
(c) 120
(d) 360

1. (c) An angle less than 90 degrees is called _____ angle.
(a) acute ✓
(b) obtuse
(c) right
(d) horizontal

1. (d) An angle of _____ is a right angle.
(a) 90 degrees ✓
(b) 80 degrees
(c) 100 degrees
(d) 180 degrees

1. (e) The length of boundary of a circle is called _____ of the circle.
(a) circumference ✓
(b) centre
(c) diameter
(d) radius

1. (f) The area covered by a closed figure is the _____ of that figure.
(a) length
(b) perimeter
(c) side
(d) area ✓

1. (g) Line of symmetry divides a figure into _____ equal parts.
(a) 5
(b) 4
(c) 3
(d) 2 ✓

2. Encircle the parallel lines from the given lines: (a), (b), (c).
Encircle the lines that never meet however far they are extended; the

parallel lines in the printed set are the ones in figure (a) (as listed in the book's answer list).

3. Differentiate right, acute and obtuse angle in the following: (a), (b), (c).
(a) Acute angle (less than 90 degrees). (b) Obtuse angle (greater than 90 degrees). (c) Right angle (equal to 90 degrees).

4. Draw the angles of the given measurements: (a) 135 degrees, (b) 70 degrees, (c) 20 degrees.
Draw a ray, place the protractor's centre point on its end point with the baseline on the ray, mark the required degree and join: (a) 135 degrees, (b) 70 degrees, (c) 20 degrees.

5. Label the circle: (a), (b), (c), (d).
(a) Diameter (the segment through the centre joining two points on the circle). (b) Centre (the middle point). (c) Circumference (the boundary of the circle). (d) Radius (the segment from the centre to the circle).

6. Find the perimeter and area of the given figures: (a), (b), (c).
(a) Perimeter 20 m, area 24 m^2. (b) Perimeter 16 cm, area 16 cm^2. (c) Perimeter 28 m, area 45 m^2.

7. Identify the symmetry in the given figures and draw lines of symmetry, where possible: (a), (b), (c), (d).
Draw a line of symmetry wherever the figure divides into two equal mirror halves; figures with no such line have no line of symmetry.

8. Complete the given figures (with respect to their line of symmetry): (a), (b), (c), (d).
Draw the mirror half on the other side of the line of symmetry so that both halves match exactly.

9. Encircle the figures which have the following properties: (a) 6 plane surfaces, (b) 1 vertex.
(a) The figures with 6 plane surfaces are the cuboid and the cube. (b) The figure with 1 vertex is the cone.

## Unit 7 - Data Handling

### Exercise 1 (book pages 165-166)

1. The minimum time for suggested sleep for children of four months to eighteen years is shown by the bar graph (scale: one horizontal square = 2

hours). (a) For the children of 6 to 12 years, suggested time duration of sleep is? (b) For which age group, the time duration of sleep is maximum and how much? (c) The sleep duration for the children of age 13 to 18 years is _____ hours less than the sleep duration for children of age 3 to 5 years. (d) For which age group the time duration for sleep is minimum and how much?

(a) For children of 6 to 12 years, the suggested sleep is 9 hours. (b) The maximum sleep is for infants: 14 hours. (c) Children of 3 to 5 years sleep 10 hours and children of 13 to 18 years sleep 7 hours, so 13 to 18 years sleep is 10 - 7 = 3 hours less. (d) The minimum sleep is for children of 13 to 18 years: 7 hours.

2. In April 2020, people from different areas and provinces of Pakistan participated in games (scale: one vertical square = 10 people). (a) The maximum number of people are from _____ and their quantity is _____. (b) The minimum number of people are from _____ and their quantity is _____. (c) From which province 30 people participated? (d) In which two provinces the people are of the same quantity? (e) What is the total number of people who participated in the games?

Reading the bars: AJ&K 5, Balochistan 10, Gilgit Baltistan 15, Islamabad 20, Khyber Pakhtunkhwa 10, Punjab 40, Sindh 30. (a) The maximum number of people are from Punjab and their quantity is 40. (b) The minimum number of people are from AJ&K and their quantity is 5. (c) 30 people participated from Sindh. (d) Balochistan and Khyber Pakhtunkhwa have the same quantity: 10 each. (e) Total = 5 + 10 + 15 + 20 + 10 + 40 + 30 = 130 people.

## Exercise 2 (book pages 169-170)

1. In a superstore, the number of customers who came for shopping for a week is shown in this line graph. (a) On what day did the maximum number of customers come and how many? (b) Is the total number of customers coming on Monday and Thursday more or less than the number of customers coming on Sunday? (c) On what day did the minimum number of customers come and how many? (d) How many customers came on Friday?

Reading the line graph: Monday 700, Tuesday 300, Wednesday 500, Thursday 400, Friday 600, Saturday 900, Sunday 1000. (a) The maximum number of customers came on Sunday: 1000. (b) Monday + Thursday = 700 + 400 = 1100, which is more than Sunday's 1000 by 100. (c) The minimum number of customers came on Tuesday: 300. (d) 600 customers came on Friday.

2. During a week of August, the maximum temperature (in centigrade) in Peshawar is shown in this line graph. (a) On what day the temperature was the highest? (b) Which two days had the same temperature and how much?

(c) What was the lowest temperature and on which day? (d) What was the temperature on Friday?

Reading the line graph: Monday 38, Tuesday 36, Wednesday 30, Thursday 35, Friday 35, Saturday 34, Sunday 32 (degrees centigrade). (a) The highest temperature was on Monday: 38 degrees C. (b) Thursday and Friday had the same temperature: 35 degrees C. (c) The lowest temperature was 30 degrees C on Wednesday. (d) The temperature on Friday was 35 degrees C.

## Exercise 3 (book pages 171-172)

1. The food components present in a boiled egg are shown in this pie chart. (a) What is the quantity of fat in the boiled egg? (b) What component is present in the most amount? (c) What is the total quantity of fat and proteins? (d) How much less is the quantity of proteins than the quantity of sodium? (e) What is the component that is present in the least amount in a boiled egg?

Reading the pie chart: Fat 5, Proteins 6, Starch 0.6, Potassium 63, Sodium 62, Cholestrol 187. (a) The quantity of fat is 5. (b) Cholestrol is present in the most amount: 187. (c) Fat + proteins = 5 + 6 = 11. (d) Sodium - proteins = 62 - 6 = 56 less. (e) Starch is present in the least amount: 0.6.

2. A survey in a village was carried out to find which cattle are bred by the people of the village. (a) Which cattle is bred the most? (b) Which cattle is bred the least? (c) Is the total number of sheep and goats more or less than the total number of donkeys and buffaloes, and how many more or less? (d) Write in the form of fraction, what part is the number of bulls to the total number of cattle?

Reading the pie chart: sheep 58, buffaloes 98, goats 100, donkeys 35, bulls 74. Total = 58 + 98 + 100 + 35 + 74 = 365. (a) Goats are bred the most: 100. (b) Donkeys are bred the least: 35. (c) Sheep + goats = 58 + 100 = 158; donkeys + buffaloes = 35 + 98 = 133; so sheep and goats are more by 158 - 133 = 25. (d) Bulls as a fraction of total cattle = 74/365.

## Review Exercise (book pages 172-174)

1. (a) It is important that in a bar graph, the _____ of each bar is same.

(a) length
(b) colour
(c) width ✓
(d) value

1. (b) Bar graphs can be of _____ types.

(a) two ✓
(b) three
(c) four
(d) five

1. (c) _____ is also called a circle graph.
(a) Pie chart
(b) Line graph ✓
(c) Vertical bar graph
(d) Horizontal bar graph

1. (d) _____ is drawn by joining the dots representing the quantity of a given value in the question.
(a) Pie chart
(b) Line graph
(c) Vertical bar graph ✓
(d) Horizontal bar graph

2. Mehwish's weight from birth to 7 years is shown in the bar graph (scale: one vertical square = 4 kg). (a) What was Mehwish's weight at the age of one year? (b) In what years Mehwish's weight was same and how much? (c) What was the age duration in which Mehwish's weight increased the most and how much? (d) What was the age duration in which Mehwish's weight increased the least and how much? (e) How much did Mehwish's weight increase in total from the age of 1 year to 7 years?
Reading the bars: first year 8 kg, second year 10 kg, third year 14 kg, fourth year 14 kg, fifth year 18 kg, sixth year 22 kg, seventh year 30 kg. (a) At one year, her weight was 8 kg. (b) Her weight was the same in the third and fourth years: 14 kg. (c) The greatest increase was from the sixth to the seventh year: 30 - 22 = 8 kg. (d) The least increase was from the first to the second year: 10 - 8 = 2 kg. (e) Total increase from 1 year to 7 years = 30 - 8 = 22 kg.

3. Ahad has planted different flowers in his garden (scale: one horizontal square = 2 flowers). (a) How many roses are there in Ahad's garden? (b) How much less are the tulips than the Gul-e-qasidi? (c) Which flowers are present in the most number in Ahad's garden and how many are they? (d) Write in fraction form the number of tulips as compared to the total number of all the flowers. (e) Which flowers are present in the least number and how many are they?
Reading the bars: Rose 4, Chrysanthemum 3, Tulip 4, Daffodil 3, Gul-e-qasidi 5, Spring flower 8. Total = 4 + 3 + 4 + 3 + 5 + 8 = 27. (a) There are 4 roses. (b) Tulips (4) are 5 - 4 = 1 less than Gul-e-qasidi (5). (c) Spring flowers are present in the most number: 8. (d) Tulips as a fraction of all flowers = 4/27. (e) Chrysanthemum and Daffodil are present in the least number: 3 each.

4. Ibrahim set up a stall of cupcakes on the spring festival of his school (line

graph, 9 in the morning to 2 in the afternoon). (a) How many cupcakes were sold in the third hour? (b) In which hour were the most cupcakes sold and what was their number? (c) In which two hours were same number of cupcakes sold? What was their number? (d) How many cupcakes were sold in the first hour?

Reading the line graph: first hour 3, second hour 6, third hour 8, fourth hour 14, fifth hour 6. (a) 8 cupcakes were sold in the third hour. (b) The most cupcakes were sold in the fourth hour: 14. (c) The second and fifth hours sold the same number: 6 each. (d) 3 cupcakes were sold in the first hour.

5. Arham asked 20 members of his family about their favourite flavours of ice-cream. (a) How many people like vanilla ice-cream? (b) Which flavour is liked the most? (c) Which flavour is liked by the same number of people and how many? (d) How much less are the number of people preferring pistachio ice-cream than the number of people preferring kulfa ice-cream? (e) There are 20 family members in total while people preferring chocolate ice-cream are 4. How will you show this number as a fraction?

Reading the pie chart: Pistachio 1, Vanilla 2, Strawberry 3, Chocolate 4, Mango 4, Kulfa 6. Total = 20. (a) 2 people like vanilla ice-cream. (b) Kulfa is liked the most: 6 people. (c) Mango and Chocolate are liked by the same number: 4 each. (d) Pistachio (1) is 6 - 1 = 5 less than Kulfa (6). (e) Chocolate as a fraction = 4/20 = 1/5.